

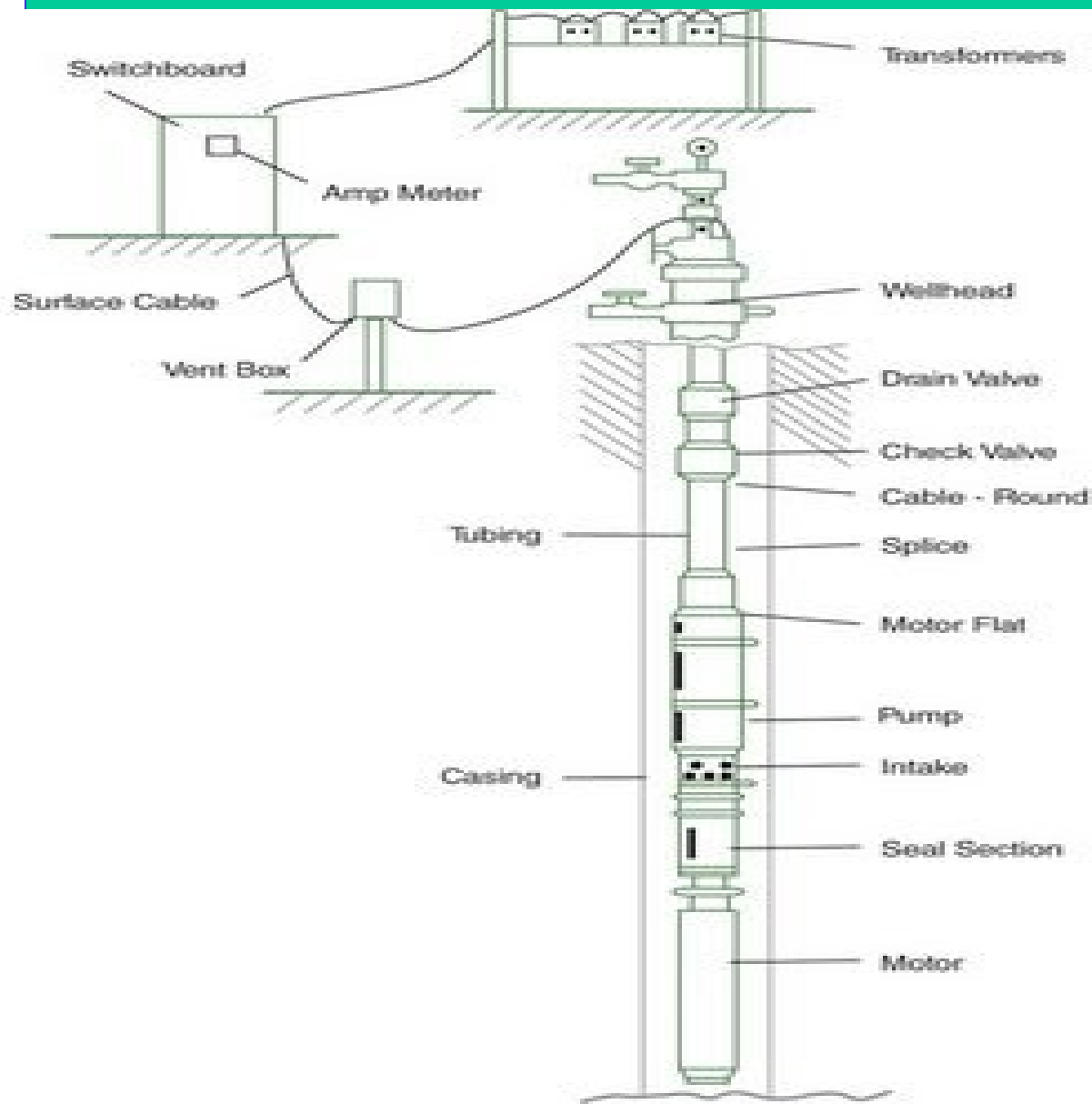
ELECTRICAL SUBMERSIBLE PUMP (ESP)

- Introduction
- Applications
- Components
 - Sub - surface
 - Surface
- Details of Sub surface components.
- Details of Surface components.
- Standard performance curves.
- Total Dynamic Head.
- Trouble shooting.
- ESP Design

Introduction

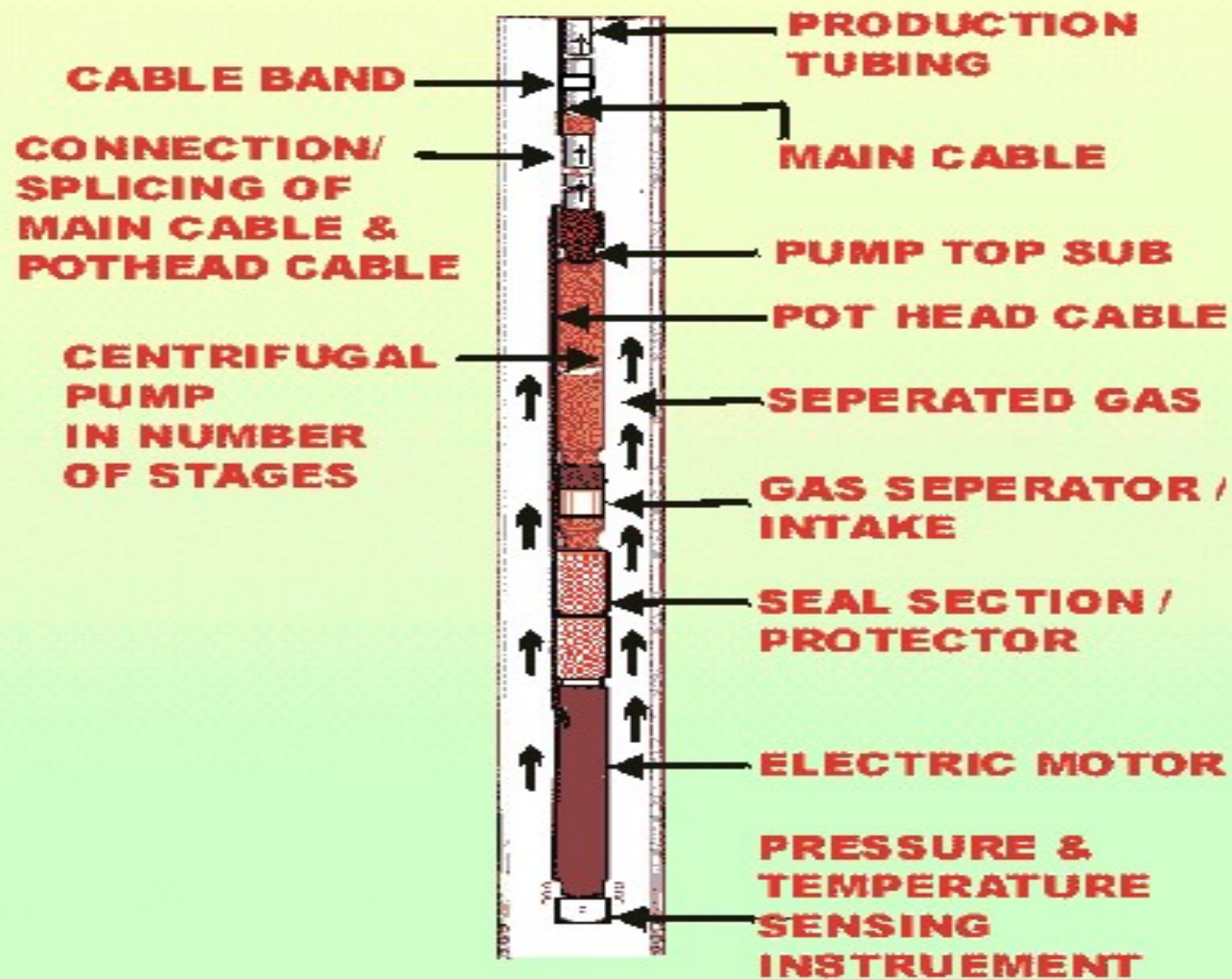
- About 15 to 20 percent of almost one million wells worldwide are pumped with some form of artificial lift employing electric submersible pumps.
- In addition, ESP systems are the fastest growing form of artificial lift pumping technology.
- They are often considered high volume and depth champions among oil field lift systems

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



COMPONENTS: SUB SURFACE

1. Electric Motor
2. Protector.
3. Pump Intake / Gas Separator
4. Multistage Centrifugal Pump
5. Pressure sensing instrument.
6. Pothead extension Cable.
7. Power cable
8. Centralizers
9. Cable glands.
10. Check valve
11. Bleeder valve (Drain valve)
12. Pump top Substitute connection
13. Lower pig tail

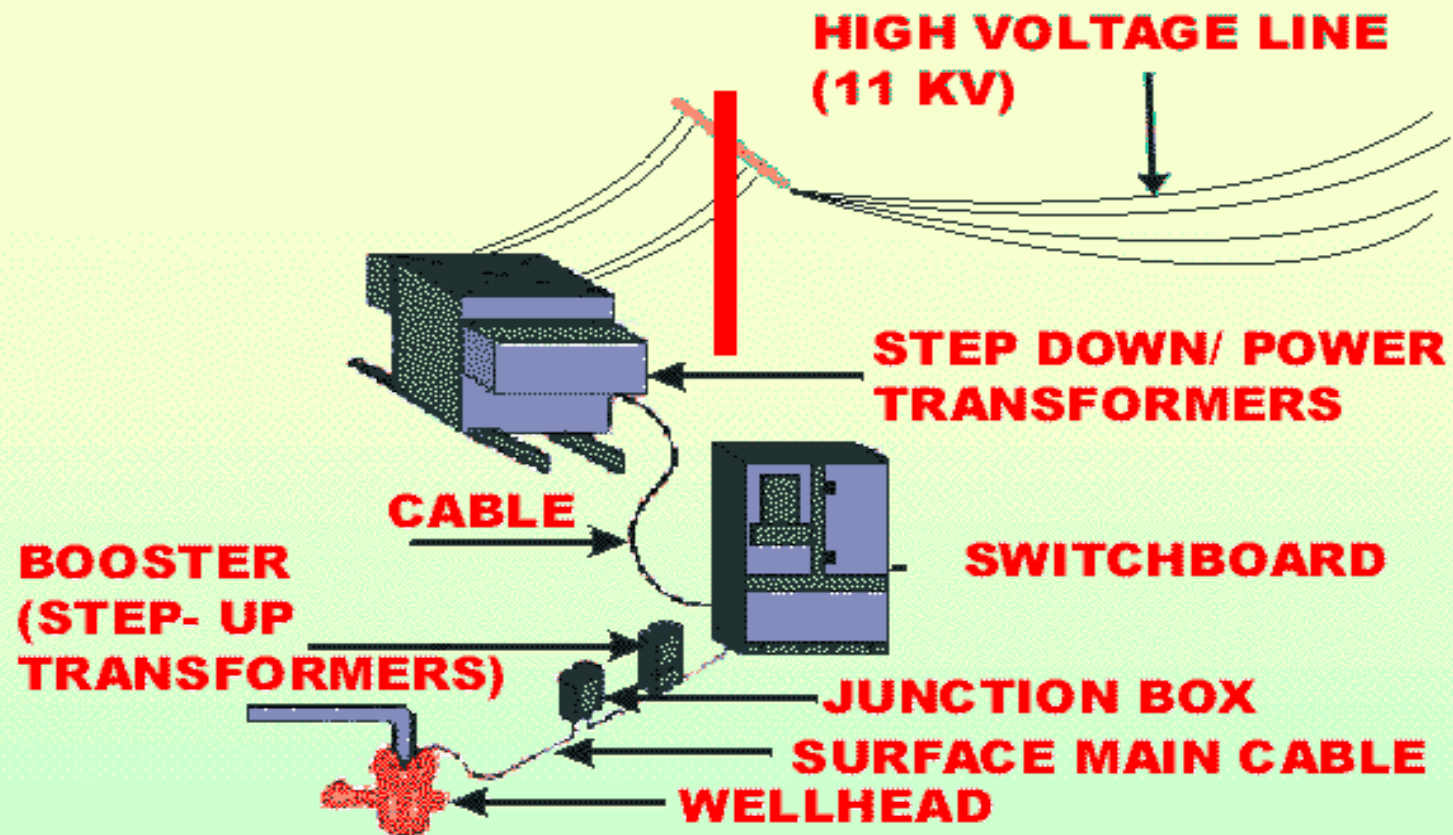


SUBSURFACE ASSEMBLY OF ESP

COMPONENTS: SURFACE

1. Well head
2. Mini mandrel.
3. Upper pig tail.
4. Surface cable
5. Junction box.
6. Booster.
7. Switch board.
8. Power transformer.

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



SURFACE SET UP FOR ESP

ELECTRIC MOTOR

It constitutes the bottommost part of the ESP assembly

- two pole-three phase squirrel cage induction type

Nominal speed

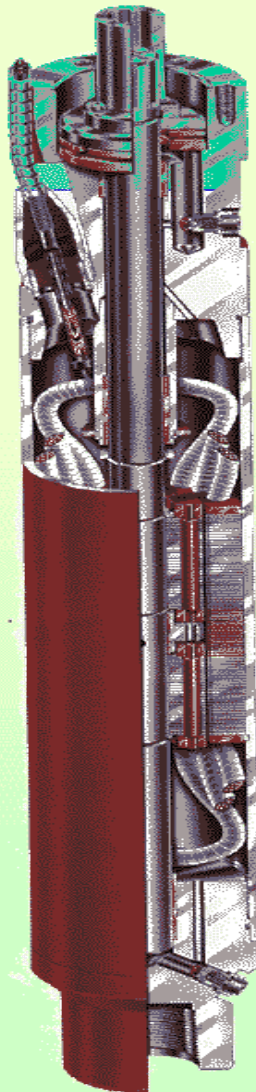
- 3500 rpm on 60 Hertz cycle
- 2915 rpm on 50 Hertz cycle

Horse power: ranges from 15 HP to 1000 HP

Diameter:

- available in different diameters
- smallest suitable for 4 1/2 casing (375 series)
- Length :ranges from 5 ft to 100 ft depending on head and volume requirements
- motor filled with highly refined mineral oil
- to provide necessary dielectric strength
- to serve as lubricant to motor bearings

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



**REDA MOTOR
(COURTSEY - REDA COMPANY)**

PROTECTOR

- placed immediately above motor
- provides seal between well fluid and motor internals
- acts as a pressure dampner/equalizer for the increasing/decreasing pressure of the mineral oil in the motor during running/stop conditions
- Two Types
 - labyrinth type
 - balloon type
 - can be of two, four, six, or eight chamber type
 - more chambers means more sealing protection

SEAL SECTION:-

- **Seal Section**—The seal section is located between the motor and the intake and performs the following functions:
- Houses the thrust bearing that carries the axial thrust developed by the pump
- Isolates and protects the motor from well fluids
- Equalizes the pressure in the wellbore with the pressure inside the motor
- Compensates for the expansion and contraction of motor oil due to internal temperature changes

Protector

- The patented modular ESP protector provides the ultimate protection available against well fluid entering the motor. There are three options for the oil reservoir chamber.
- The labyrinth section simply uses the differential weight of the fluid to create a barrier and a maze for the wellbore fluid to travel through before it enters the motor. These were the first protectors used and are still very popular in vertical wells.

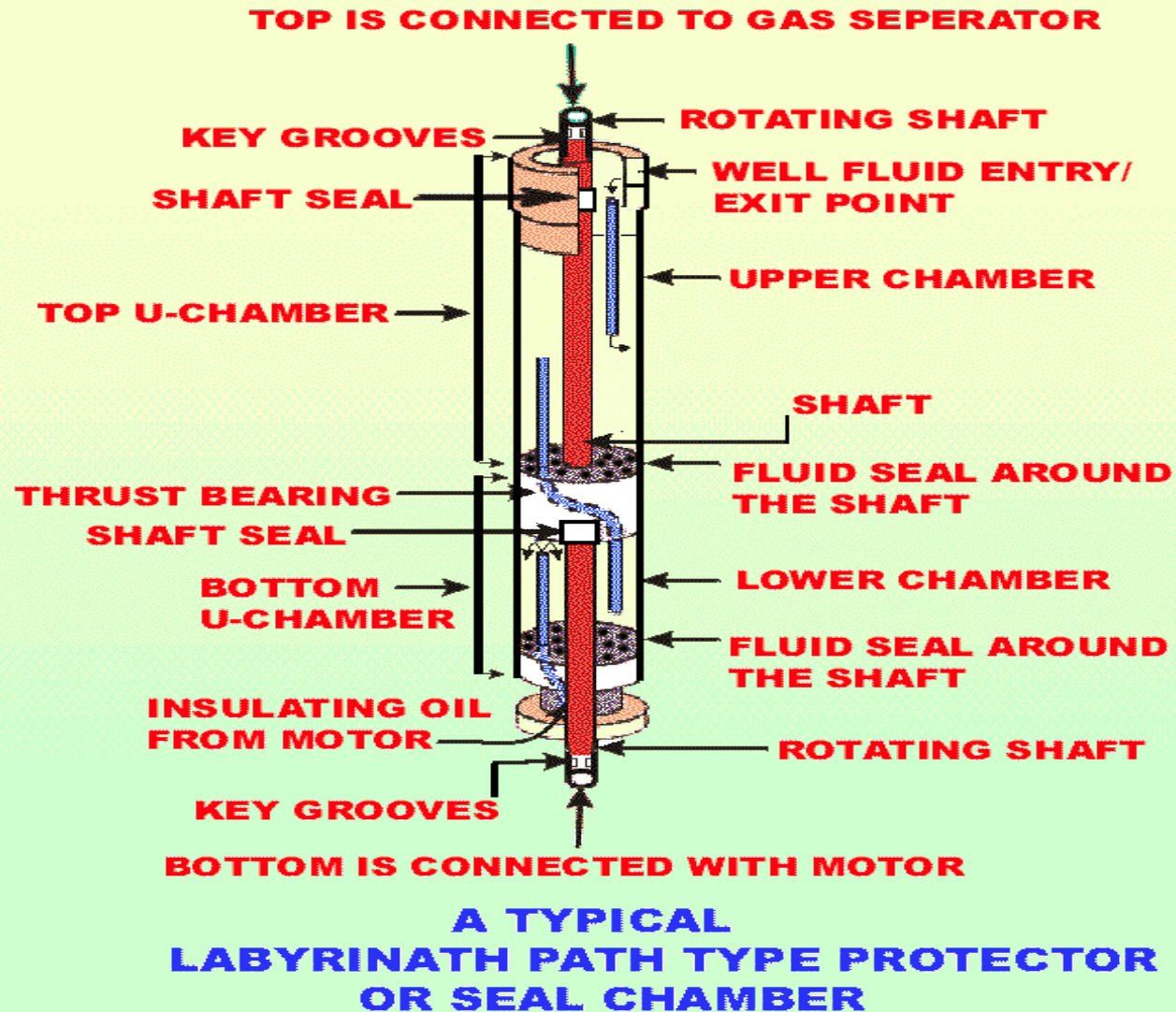
Protector

- The positive seal protector, or elastomer bag, provides a barrier between the well fluids and the motor oil to allow operation in gaseous or deviated wells. The bags come in different materials depending on the application.
- The metal bellows is also a positive seal to well fluids and provides an impervious barrier to H_2S , which tends to permeate elastomers and enter the motor. The bellows is also rated for high-temperature applications up to 475 degF (246 degC).

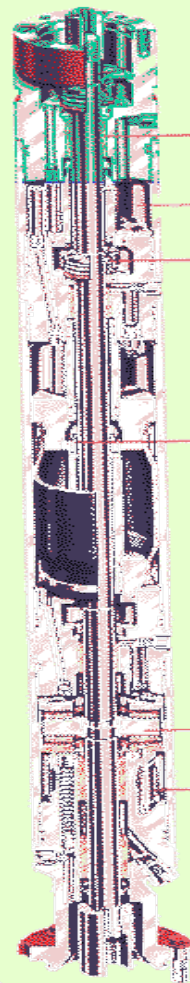
Protector

- In addition to the function of a seal, the ESP protector also serves as an oil reservoir for the motor, balances the internal pressure of the well fluid, and carries the thrust load of the pump.

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



**EXTRA SHAFT
BUSHINGS PROVIDE
GREATER SHAFT
SUPPORT**

**LONGER, HIGH
CAPACITY LABYRINTH
CHAMBERS**

**EXTRA SHAFT SEALS
IN MOST MODELS
FOR IMPROVED
MOTOR PROTECTION**

**SHEDDER DEFLECTS
ANY WELL FLUID
MIGRATING PAST
SHAFT SEAL INTO THE
ELASTOMER BAG**

**HIGH PERFORMANCE
THRUST BEARING**

**FILTER SCREEN
PREVENTS DEBRIS
FROM ENTERING
THE MOTOR**

**REDA MODULAR PROTECTOR
(COMBINATION OF LABYRINTH
AND POSITIVE SEAL)**

(COURTSEY - REDA COMPANY)

PUMP INTAKE/ GAS SEPERATOR

- Connected between protector and pump section
- provides suction to the pump

Two Types

straight pump intake:-provides suction to the pump with out separating gas

gas separator intake: -separates gas and allows only liquid to the pump suction

--Types in Gas Separator

Poor Boy or Reverse flow separator

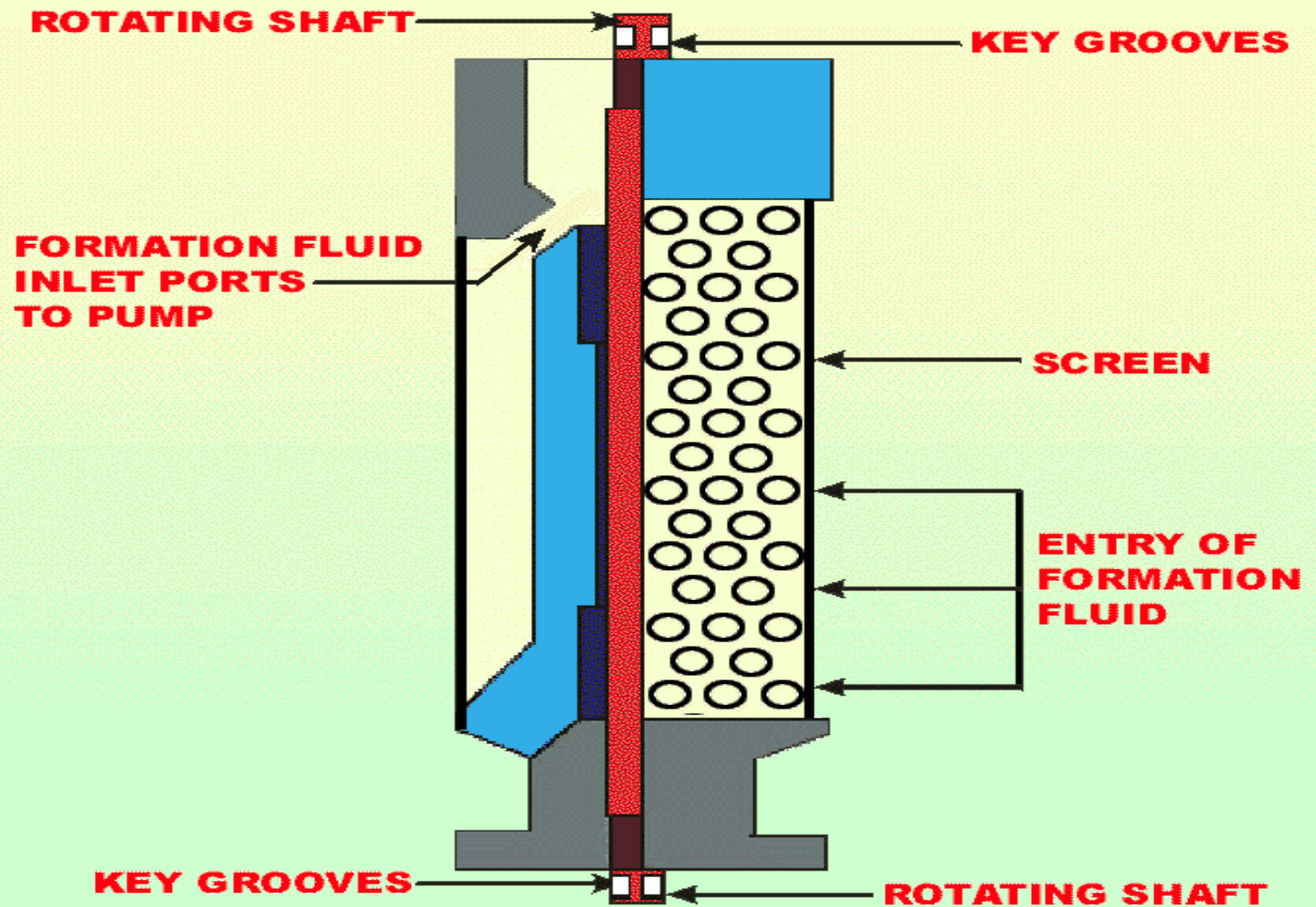
routes well fluid 180 degrees downwards so that gas escapes upward in to annulus and only liquid enters through pull tube

Rotary gas separator

works on centrifugal action. Keeps the heavier fluid (liquid) to

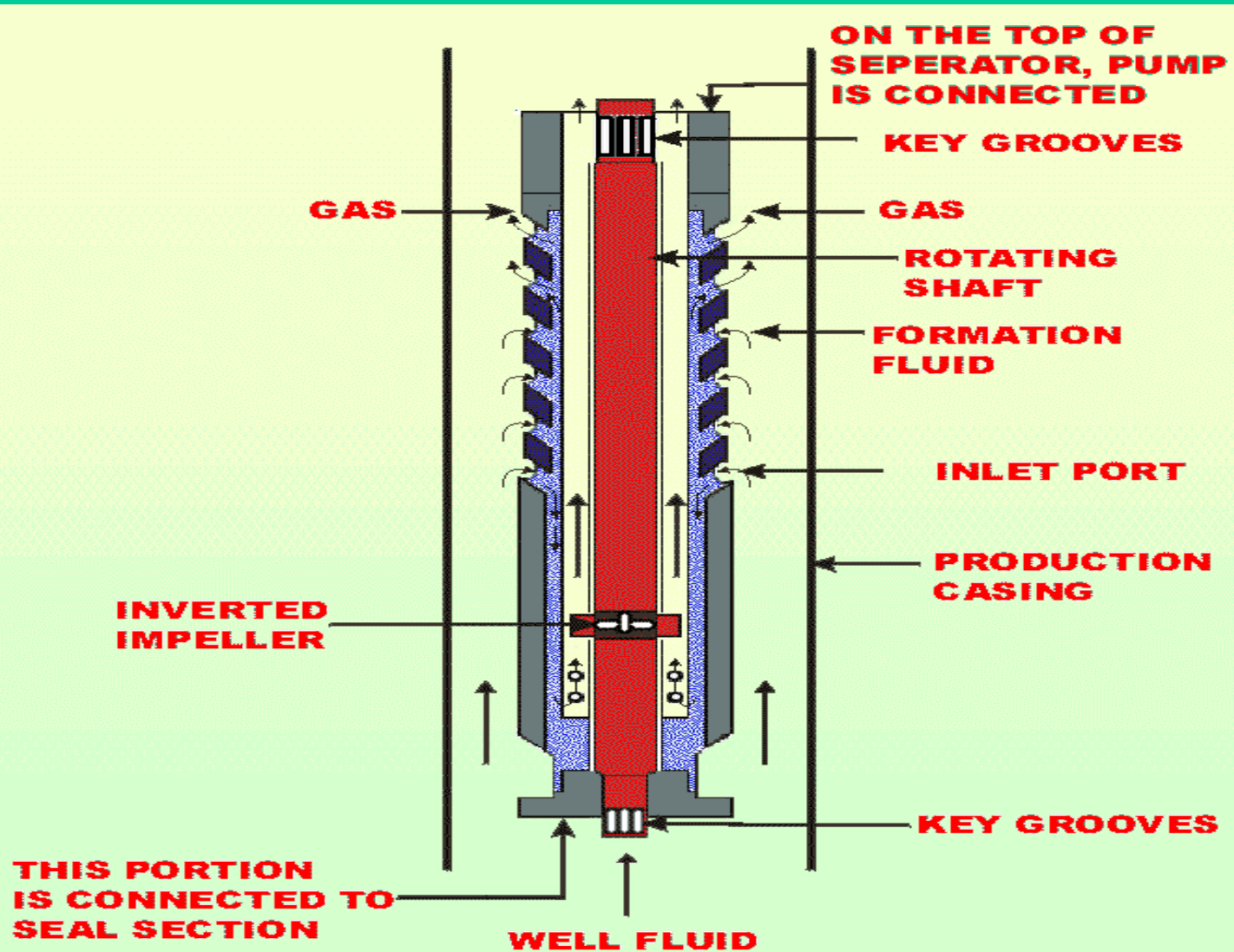
the periphery where as lighter fluid (gas) flows through the center in to annulus via a flow devoured

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



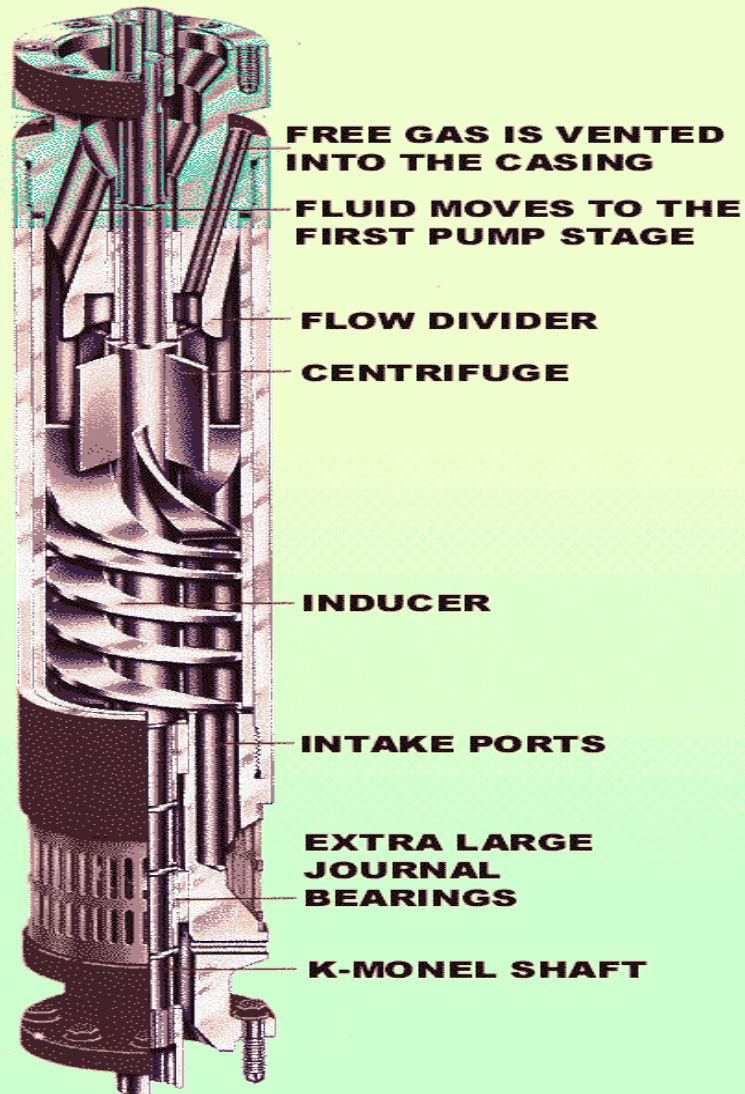
**A TYPICAL STANDARD INTAKE
(WITHOUT GAS SEPERATOR)**

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



A TYPICAL REVERSE FLOW GAS SEPERATOR

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



**REDA ROTARY GAS SEPERATOR
(COURTSEY - REDA COMPANY)**

Vortex gas separator



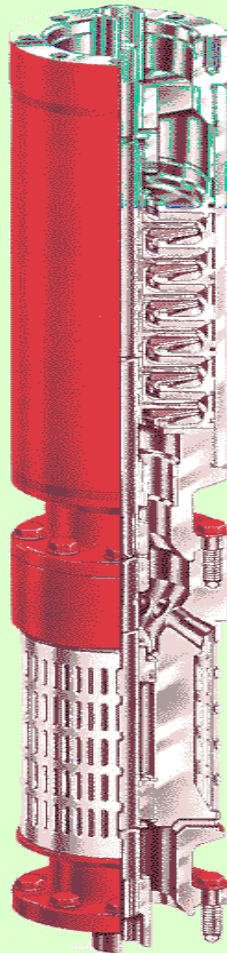
MULTISTAGE CENTRIFUGAL PUMP

- Nothing but several centrifugal pumps in series
- OD and impeller design determines the production rate
- no. of stages determines the head developing capability
- each stage consists of one impeller and diffuser
- Two Types of Impeller Diffusers
 - Floating or Balanced type: Impeller floats up and down axially along the shaft depending on the designed and actual fluid production rates. Can tolerate slight fluctuations in the production rates
 - Fixed Impeller type : Impeller is fixed to the shaft. Does not move up and down.
suitable for pumping very high volumes

MULTISTAGE CENTRIFUGAL PUMP(CONTD)

- Each pump housing may contain 54,74,99 or 151 stages
 - Two or more housings are connected together to create necessary no. of stages
 - length of each pump housing varies from 2.1 ft to 14.8 ft
- ⇒ Metallurgy :
 - housing -seam less, heavy walled, low carbon steel tubing
 - shaft -K-Monel
 - Impellers - Ni-resistant, Ryton, or Bronze
 - Diffusers - Ni-resistant

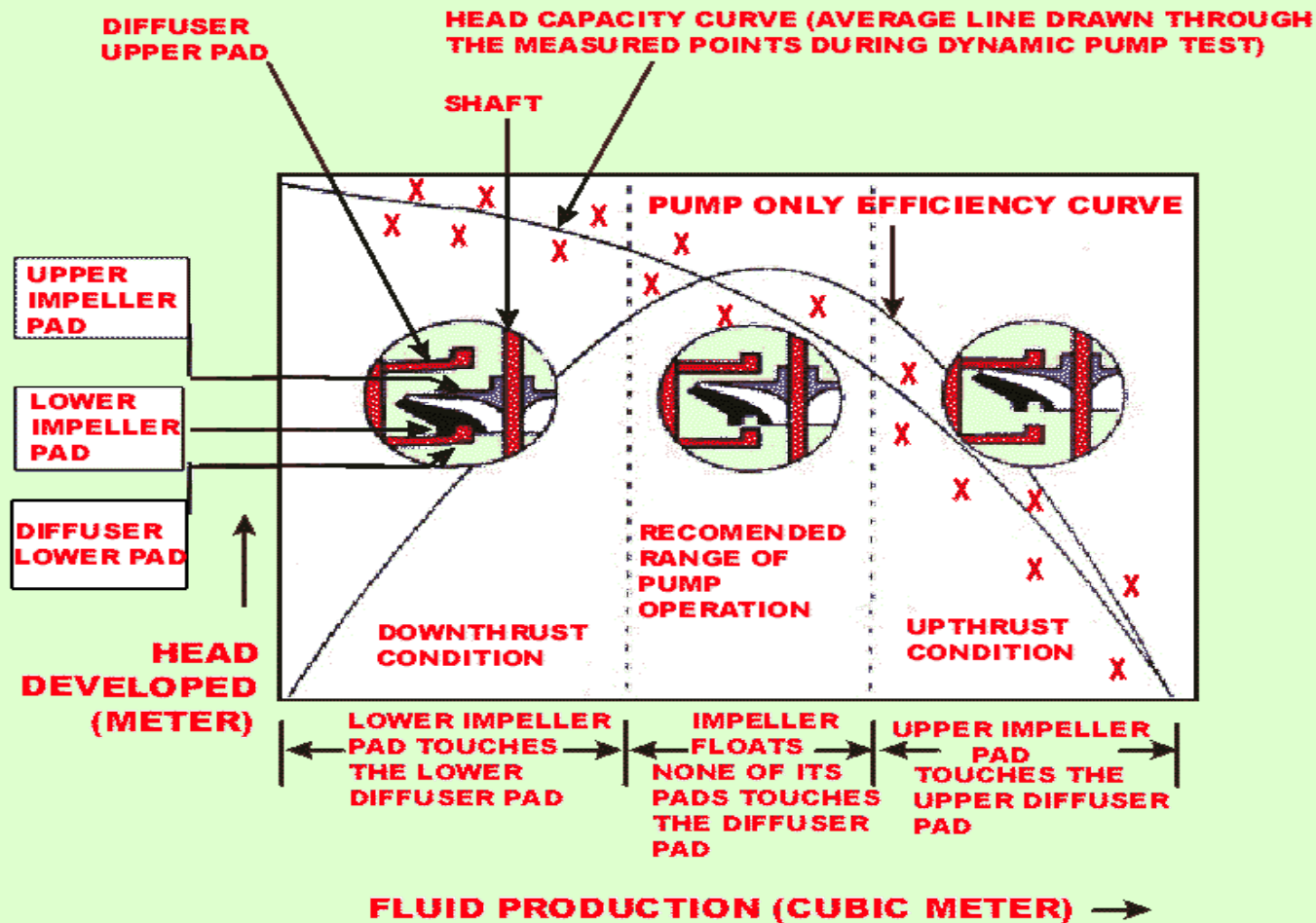
ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



**NEW STAGE DESIGNS INCREASE
OPERATING EFFICIENCIES UPTO 6%**

**REDA HIGH EFFICIENCY PUMP
(COURTSEY - REDA COMPANY)**

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



DRAWAL OF HEAD CAPACITY & PUMP EFFICIENCY CURVE OF A TYPICAL ESP

PRESSURE SENSING ELEMENT (OPTIONAL)

- fitted below the motor
- records the pressure and temperature of the fluid at the pump depth
- surface recorder unit connected to the downhole sensor
through the same power cable and motor windings

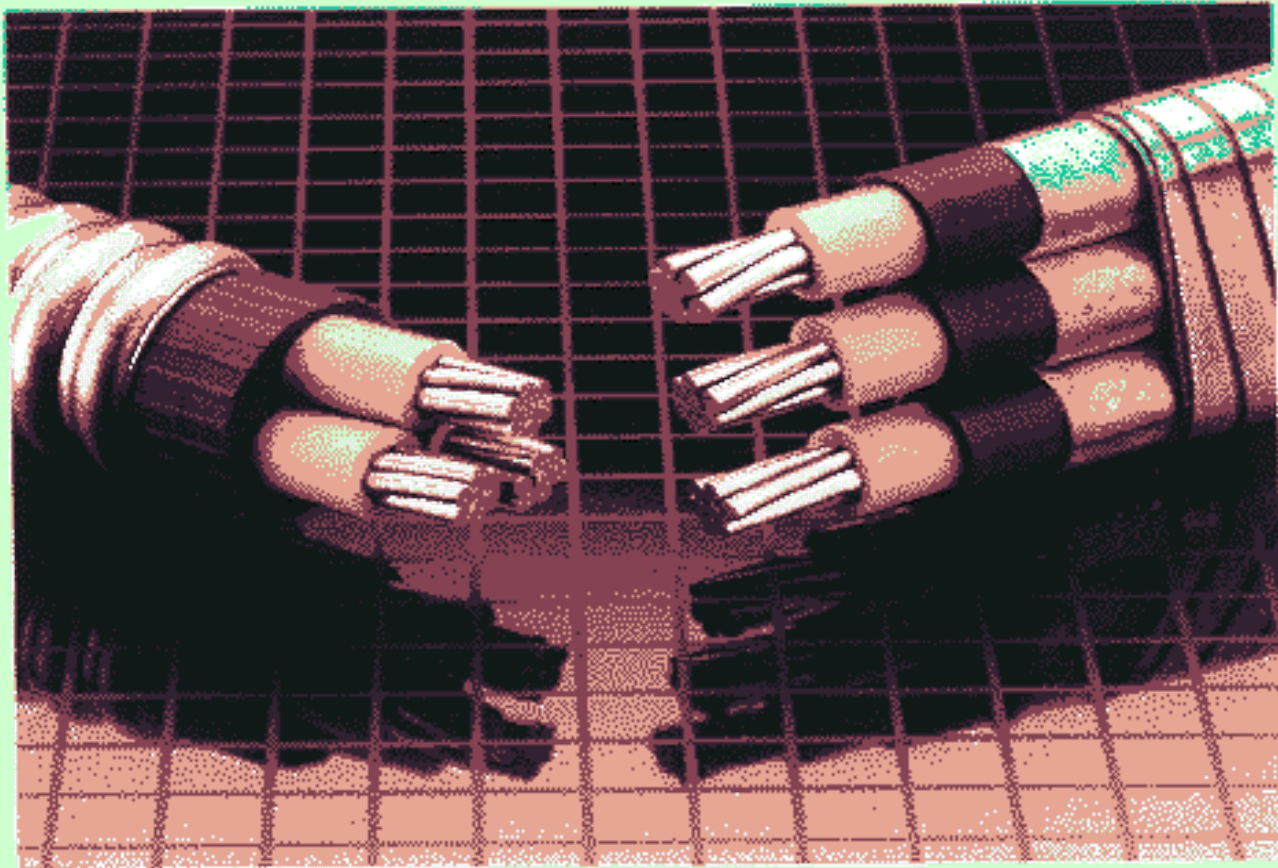
POTHEAD EXTENSION POWER CABLE

- connects the motor windings to the main power cable
- very sensitive part of ESP system
- Two Types :
- Plug in type
 - similar to three pin plug with 'O' ring fitted for fluid seal
 - connected to the motor three pin hole and bolted
 - even though easy to connect, chances of developing hairline cracks while fitting, are more, thus leading to insulation failures
- Tape in type
 - similar to taped joint between two cables
 - needs well trained technician for jointing
 - highly reliable, once good splicing job is ensured

POWER CABLE

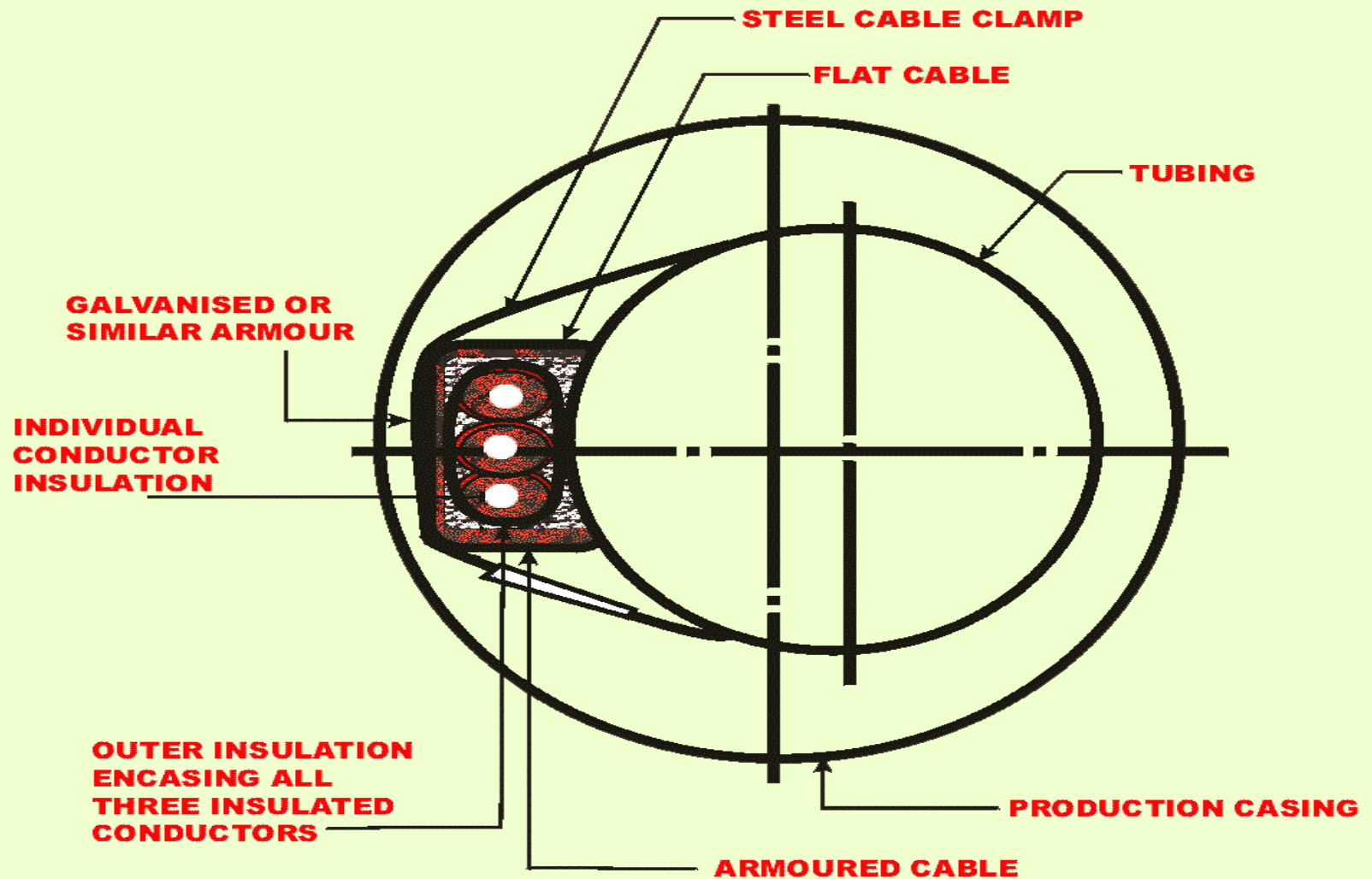
- Supplies power from surface to the down hole motor thickness varies from # 1 AWG TO# 6AWG.
- More the gauge number, less thicker and less current carrying capacity even though armored and insulated, most susceptible component while lowering or pulling out the pump by abrasion between casing- tubing walls
- Two types
 - Round cable: round in shape, three conductors placed side by side at 120 degrees to each other because of large diameter ,clearance becomes less specially at the tubing coupling, hence more chances of abrasion gripping to the tubing is less because of less area of contact
 - Flat cable: flat in nature, three conductors placed side by side more commonly used because of i) firm gripping possible, ii) availability of more clearance

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



**REDA ROUND AND FLAT CABLE
(COURTSEY - REDA COMPANY)**

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



ESP DOWNHOLE FLAT CABLE IN CLAMPED ON POSITION

CENTRALIZERS

--one below the motor and one above the pump will keep the assembly in the center of the casing for better cooling affect

CABLE BANDS

- though a small item, one of the vital component
- attaches the cable rigidly with the tubing
- about 3 to 4 cable bands are required per stand
- breaking down cable bands leads to serious problems
 - like i) shifting of cable weight to the next lower band, which may lead to the failure of that band and the process continues, ii) this leads to the coiling of cable while pulling out tubing for any repair, warranting costly cable replacement iii) may finally lead to the jamming of tubing iv) sometimes responsible for even abandoning the well
- one time use only, replaced for re-run of ESP

CHECK VALVE (OPTIONAL)

- a flapper disc type non-return valve
- installed just above the pump
- keeps tubing always full
- prevents reverse flow of fluid, thus preventing reverse rotation of the impeller, which sometimes may become dangerous to the motor.

BLEEDER(DRAIN) VALVE(OPTIONAL)

- installed above the check valve
- useful for draining out liquid from the tubing during pulling out job otherwise oil/water may be splashing on the derrick floor
- can be opened before ESP pullout, by dropping a heavy rod from top.
- may warrant tubing p/o if accidentally broke open

PUMP TO SUB

- connects pump to the tubing
- flanged on the pump side and box threaded on the tubing side

LOWER PIG TAIL

- required when “SEABOARD” or equivalent type of wellhead is used
- it is a small length of main cable , spliced with the main cable before wellhead is installed and other end is connected with the electrical ‘minimandrel’

WELL HEAD

ESP well heads are different from normal well heads

⇒ Two Types

⇒ 'Hercules' equivalent make

- sub surface cable protrudes to the surface through the well head, Sealing is achieved by rubber packing around the cable. Requires to subdue the well for change of packing in case of leakage.
- can with stand pressures upto 1500 psi
- to be specified for round or flat cable as well as AWG of the cable

WELL HEAD(CONTD)

'Seaboard' or equivalent make

- sub surface cable is joined to one end of the lower pig tail and other end of the lower pig tail is in turn connected to the minimandrel. Minimandrel is screwed in to the wellhead with necessary 'O' ring
- can withstand much higher pressures of up to 3000 psi

MINI MANDREL

- ☞ three copper conductors of required size enclosed in very good non-conducting solid material in cylindrical shape
- ☞ screwed in to the well head (Sea Board type)
- ☞ ends coupled with lower and upper pig tails

UPPER PIG TAIL

- similar to the lower pig tail
- one end coupled with mini mandrel and other end joined with surface cable

SURFACE CABLE

Similar to power cable

- connects either to switch board or junction box
- preferable to lay through tubing from wellhead to switch board to avoid any physical damage

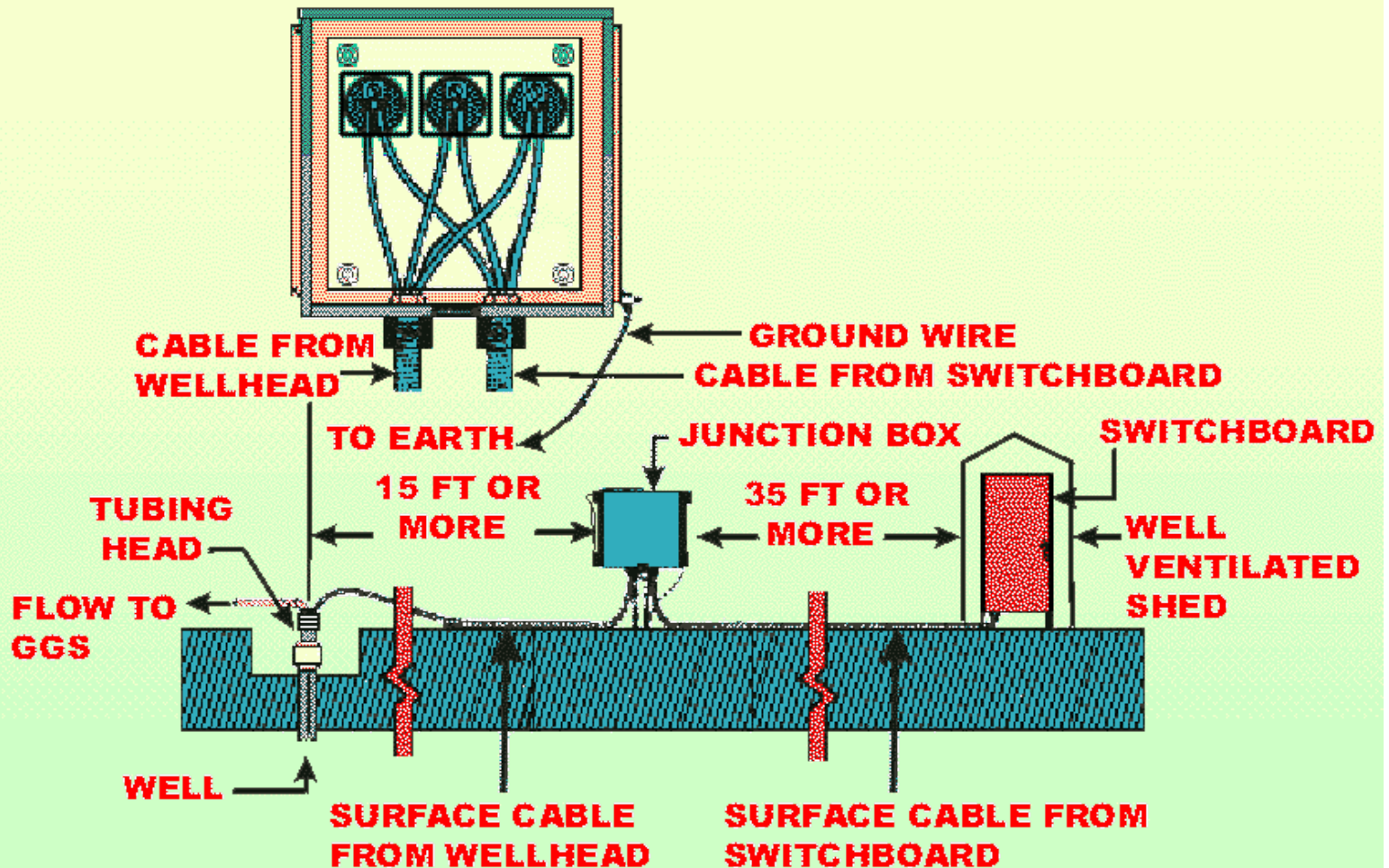
JUNCTION BOX

- required when 'Hercules' or equivalent make well head is used
- to connect the sub surface cable and surface cable at a safe distance from the wellhead
- it is a well ventilated box with a provision for venting the gas which may get carried through the power cable

BOOSTER

- required to boost the surface voltage according to the requirement of rated downhole voltage
- connected preferably between the junction box and the switch board

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



JUNCTION BOX SHALL BE WELL VENTILATED & PROPERLY GROUNDED

JUNCTION BOX INSTALLATION

SWITCHBOARD

- standard switchboards are weatherproof but not necessarily flameproof
- available in voltage ranges from 440 volts to 4900 volts
- contains recording ammeter, fused disconnects, underload and overload protections, signal lights, timers for intermittent auto start
- shuts down the unit in both underload and overload conditions
- can be set to auto re start with time delay (with timer) in case of underload trip
- to be manually re started in case of overload trip

POWER TRANSFORMER

- steps down the voltage to the required level
- available in different kW ranges

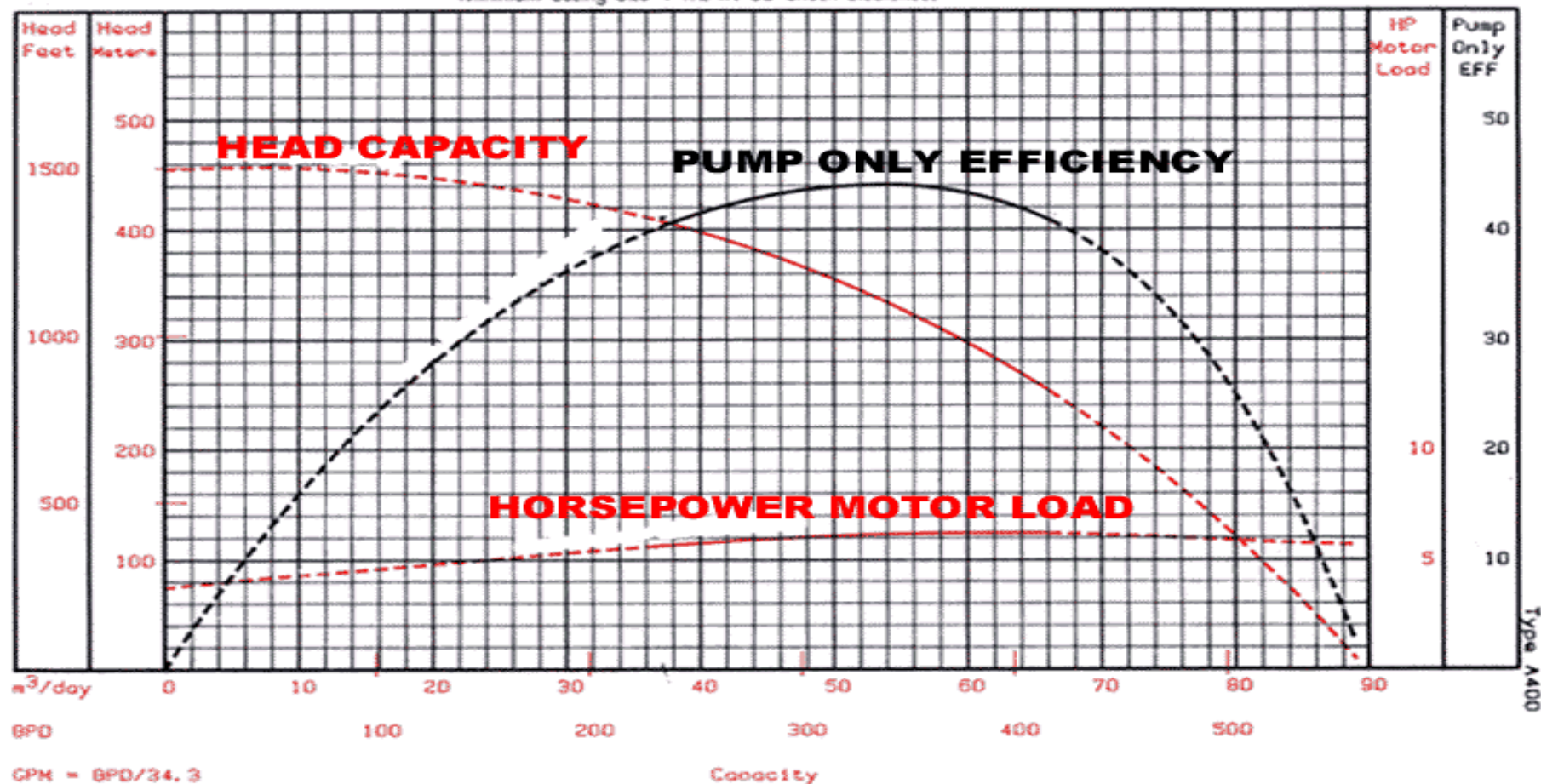
STANDARD PERFORMANCE CURVES

- indicates the dynamic flow condition of ESP
- different for different HP and size of the pump
- abscissa indicates pumping capacity in std.bbl/d or m³/d
- ordinate indicates liquid head generated in ft or Mts.
- is plotted with either 100 stages of pump or single stage depending on the manufacturer
- head will be maximum at very low or zero rates and conversely head will be zero at the maximum rate
- the space between the maximum and minimum lines is called the operating range of that ESP
- using this chart, HP and no. of stages required can be calculated for any given head and rate

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve
100 Stage — A400 — 50 Hz
338 Series — 2917 RPM

Minimum Casing Size 4 1/2 IN OD Check Clearances



(COURTSEY - REDA COMPANY)

Type A400

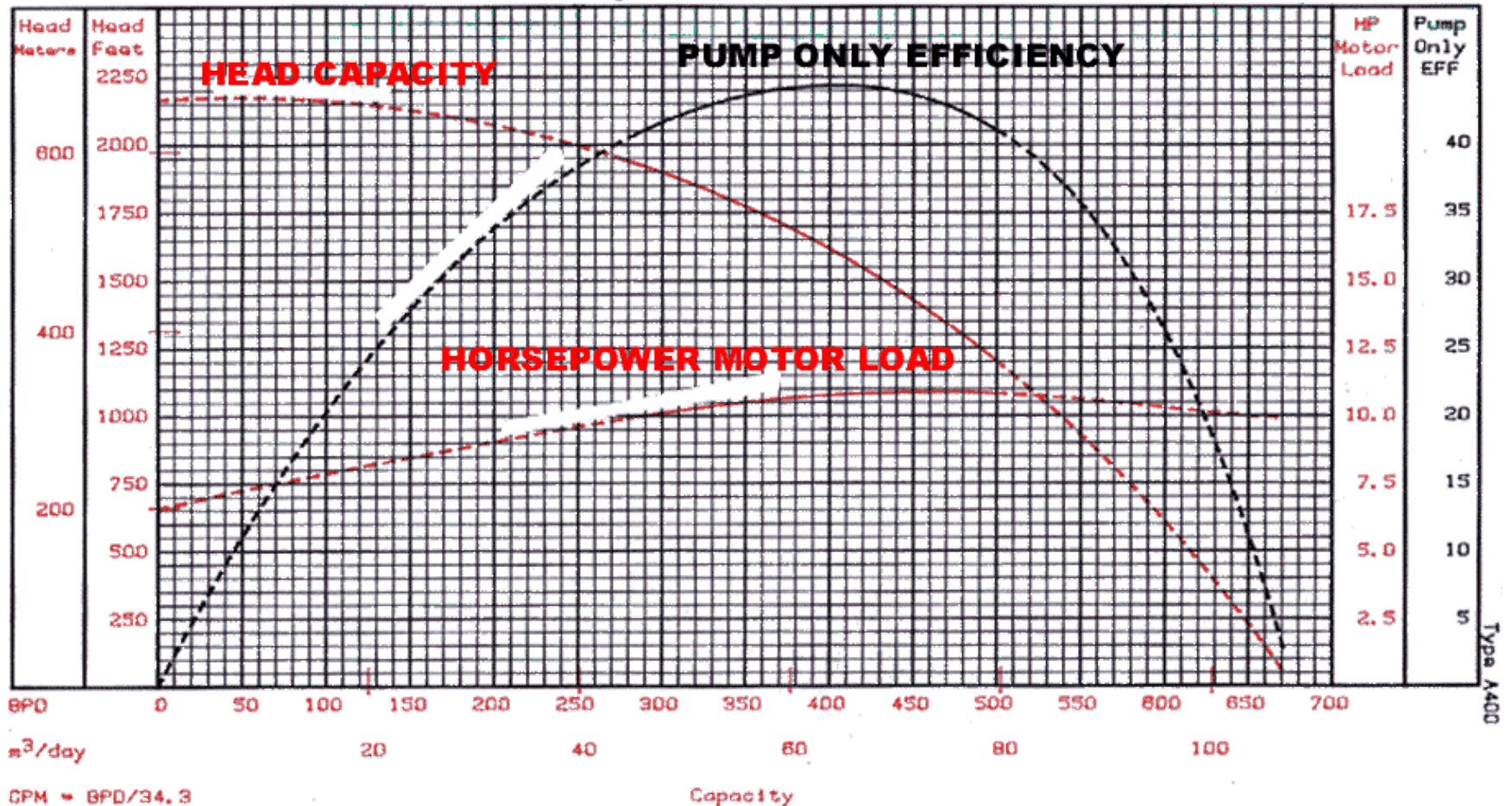
Performance Curves

- Describe the performance of a particular impeller(stage type)

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve 100 Stage — A400 — 60 Hz 338 Series — 3500 RPM

Minimum Casing Size 4 1/2 IN OD Check Clearances



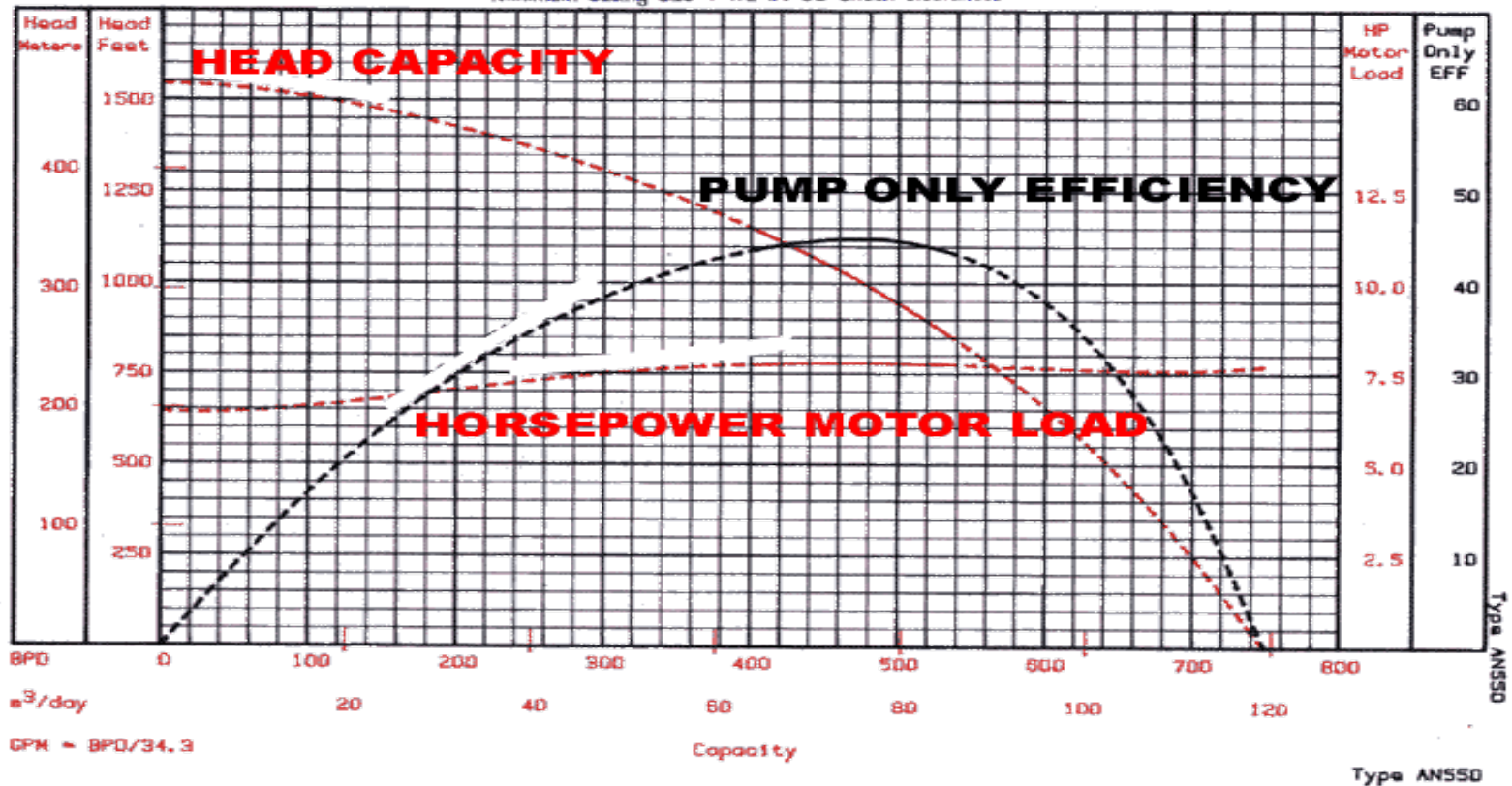
(COURTSEY - REDA COMPANY)

Type A400

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve
100 Stage — AN550 — 50 Hz
338 Series — 2917 RPM

Minimum Casing Size 4 1/2 IN OD Check Clearances

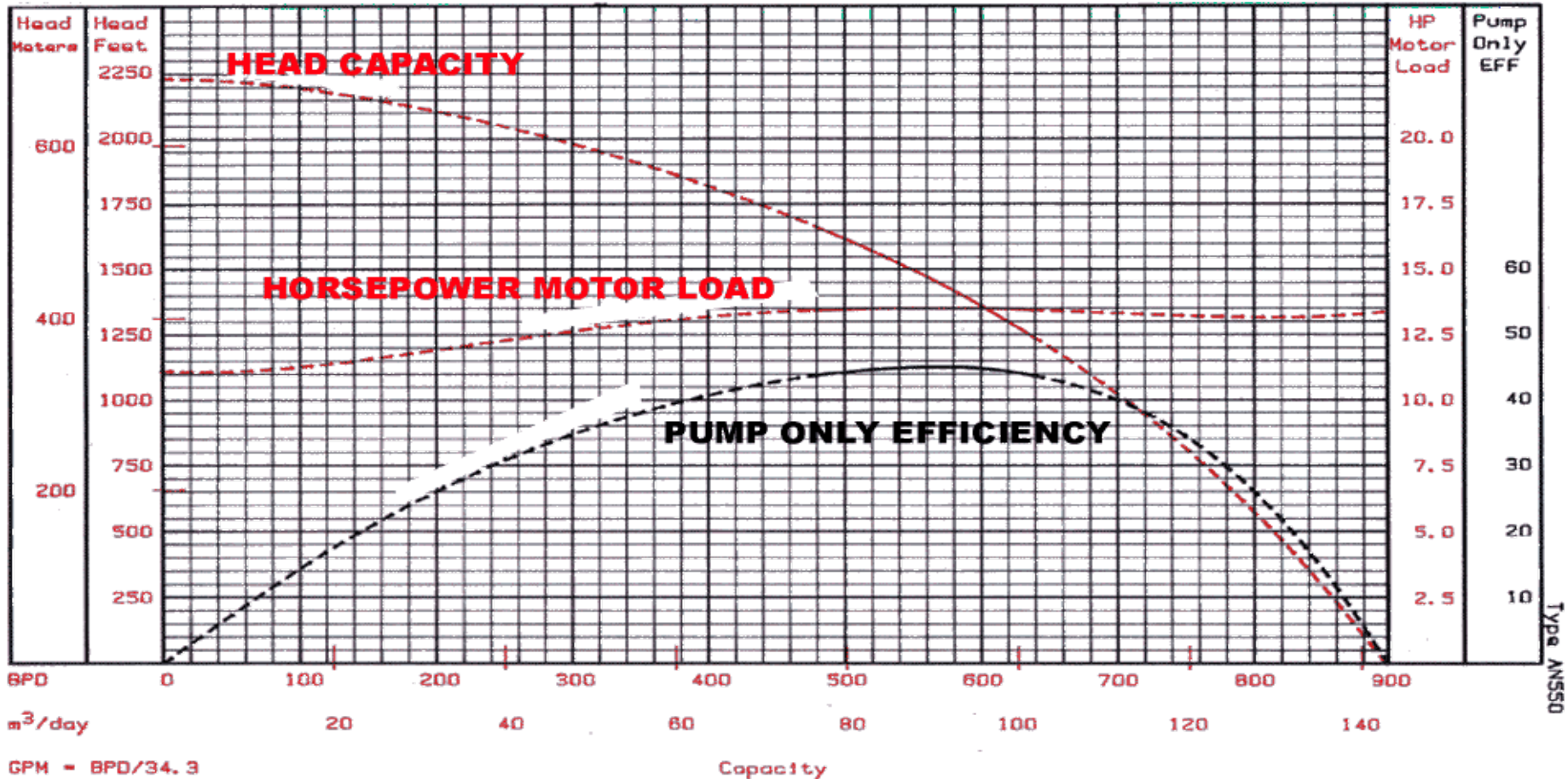


(COURTSEY REDA COMPANY)

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve 100 Stage — AN550 — 60 Hz 338 Series — 3500 RPM

Minimum Casing Size 4 1/2 IN OD Check Clearances

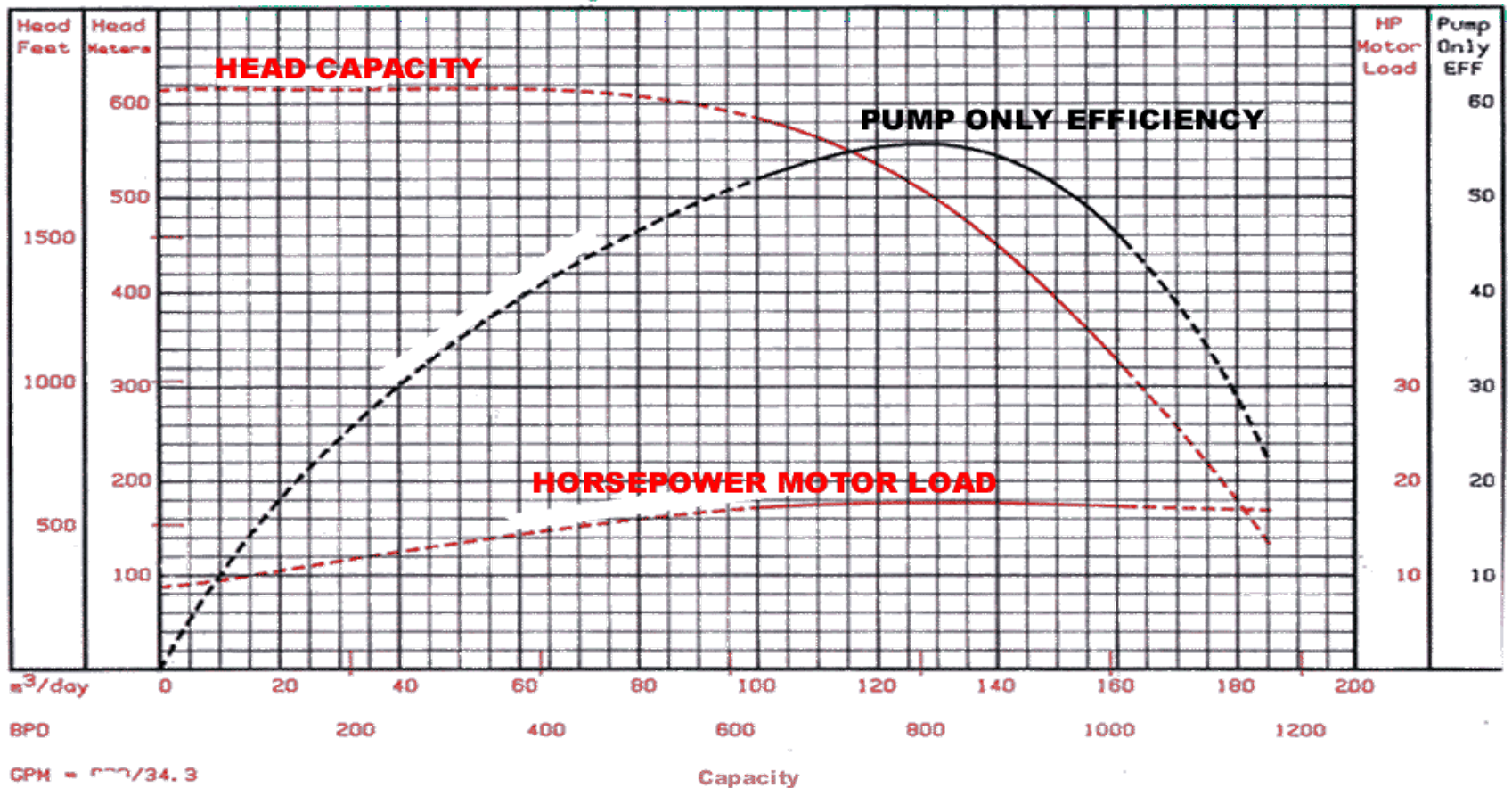


(COURTESY- REDA COMPANY)

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve
100 Stage — DN1000 — 50 Hz
400 Series — 2917 RPM

Minimum Casing Size 5 1/2 IN OD Check Clearances



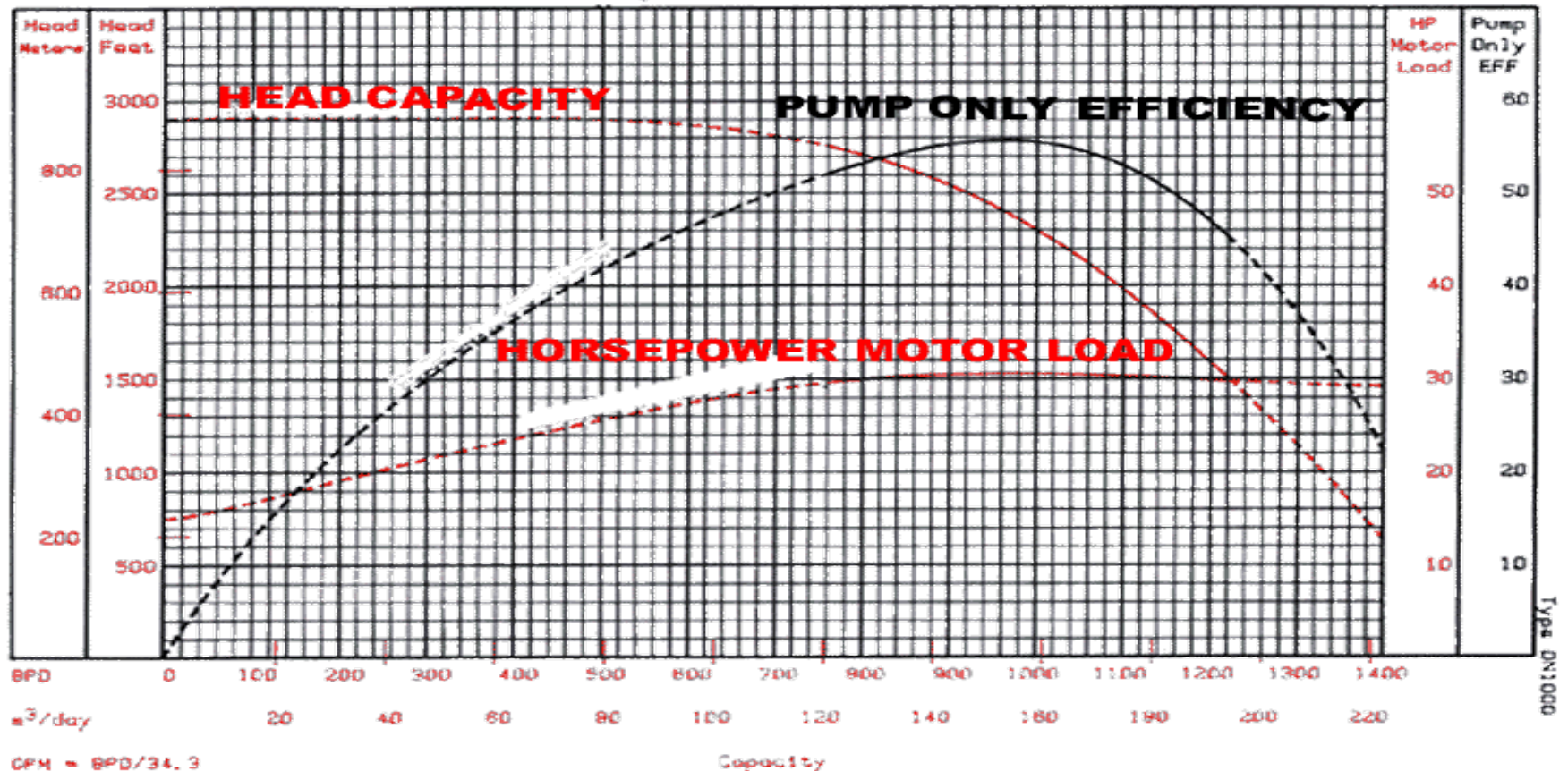
pe DN1000

(COURTESY REDA COMPANY)

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM

Reda Pump Performance Curve
100 Stage — DN1000 — 60 Hz
400 Series — 3500 RPM

Minimum Casing Size 5 1/2 IN OD Check Clearances

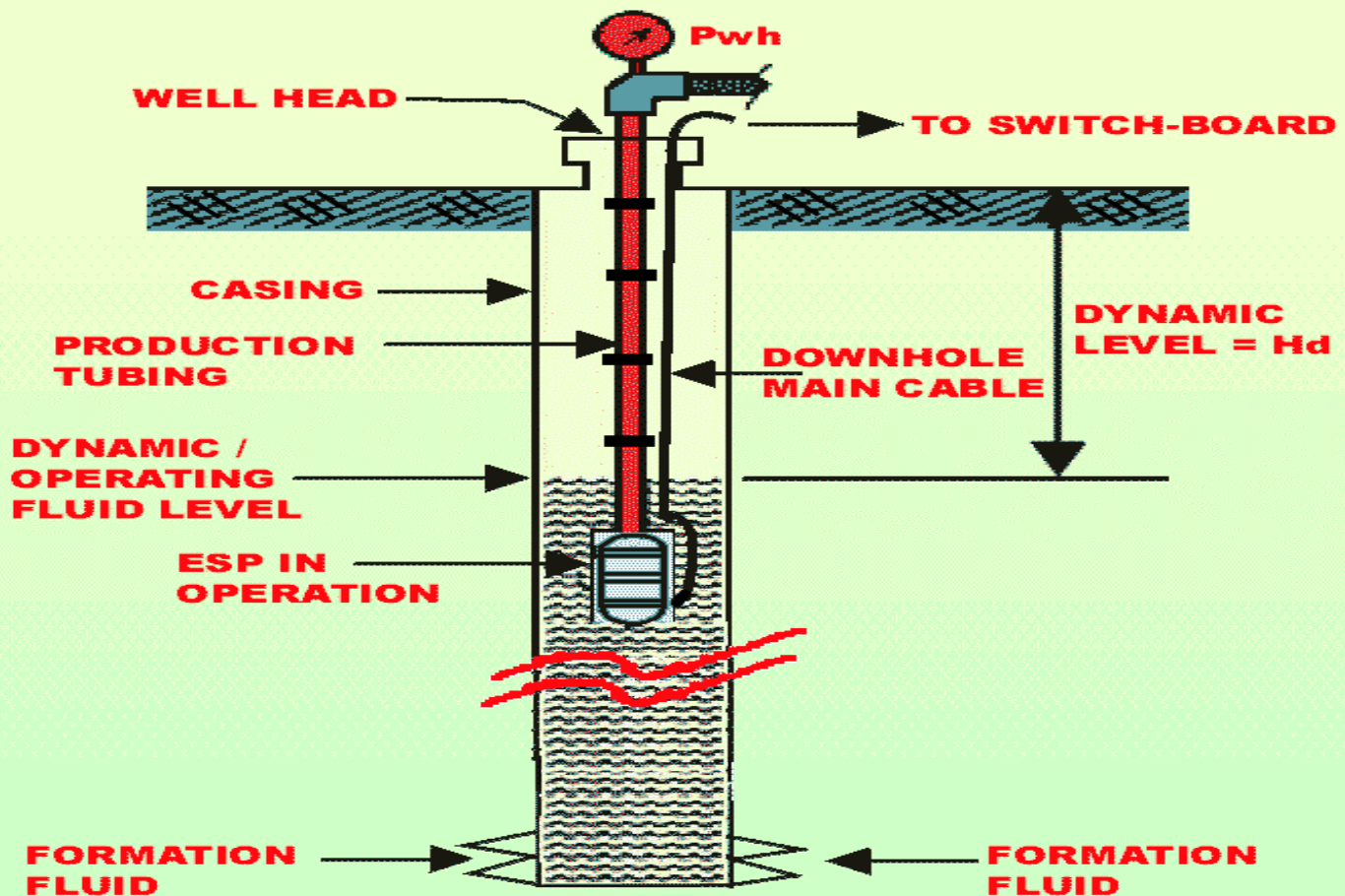


(COURTESY REDA COMPANY)

TOTAL DYNAMIC HEAD (TDH)

- useful concept for calculating the total number of stages of a centrifugal pump
- TDH includes i) the friction losses in the tubing and the surface line, ii) tubing pressure against which pumping is to be done, iii) difference in elevation of the dynamic level and the surface, and iv) any losses due to valves etc in the flowline

ELECTRICAL SUBMERSIBLE PUMPING (ESP) SYSTEM



$$\text{TOTAL DYNAMIC HEAD} = P_{wh} + H_d$$

CALCULATION OF DYNAMIC HEAD FOR ESP

TROUBLE SHOOTING OF ESP

- ☞ In trouble shooting any malfunctioning ESP, following data will be very helpful in analyzing the problem and thus avoid costly workover jobs
- Rate of oil production
- Water cut
- Run life of ESP unit
- Dynamic and static fluid levels
- GOR/GLR
- Pump setting depth
- Sand cut/ corrosive fluid etc.
- How many times ESP has been serviced and for what reasons
- Reservoir pressure/Reservoir drive mechanism/ rate of fall of static pressure
- Bottom hole temperature

TROUBLE SHOOTING OF ESP(CONTD)

- Recording ammeter, which continuously records the current drawn by the pump, is also a very useful tool in analyzing the problem of a faulty/underperforming ESP.
- Many problems like, electrical supply fluctuation, gas locking, fluid pump-off conditions, false starts, underload /overload conditions can be diagnosed from the recorded ammeter charts.

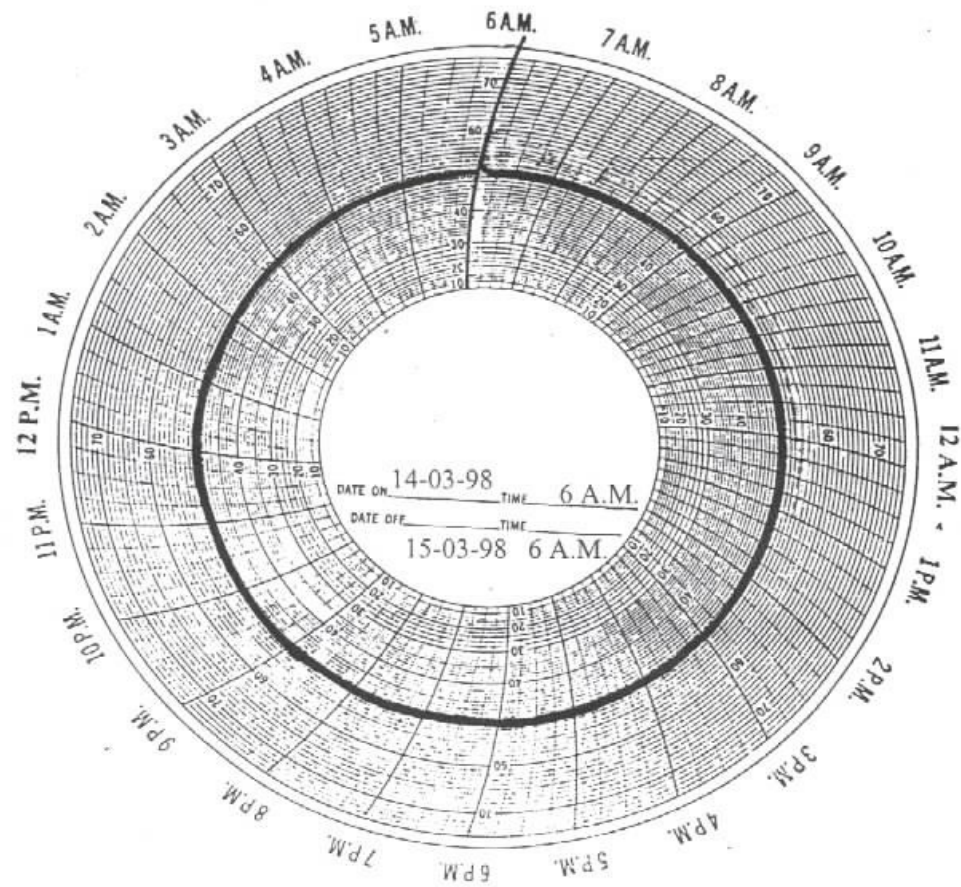


fig. 1

Pump is in normal & smooth operation, since pumping operation continues with almost negligible fluctuation in amperage.

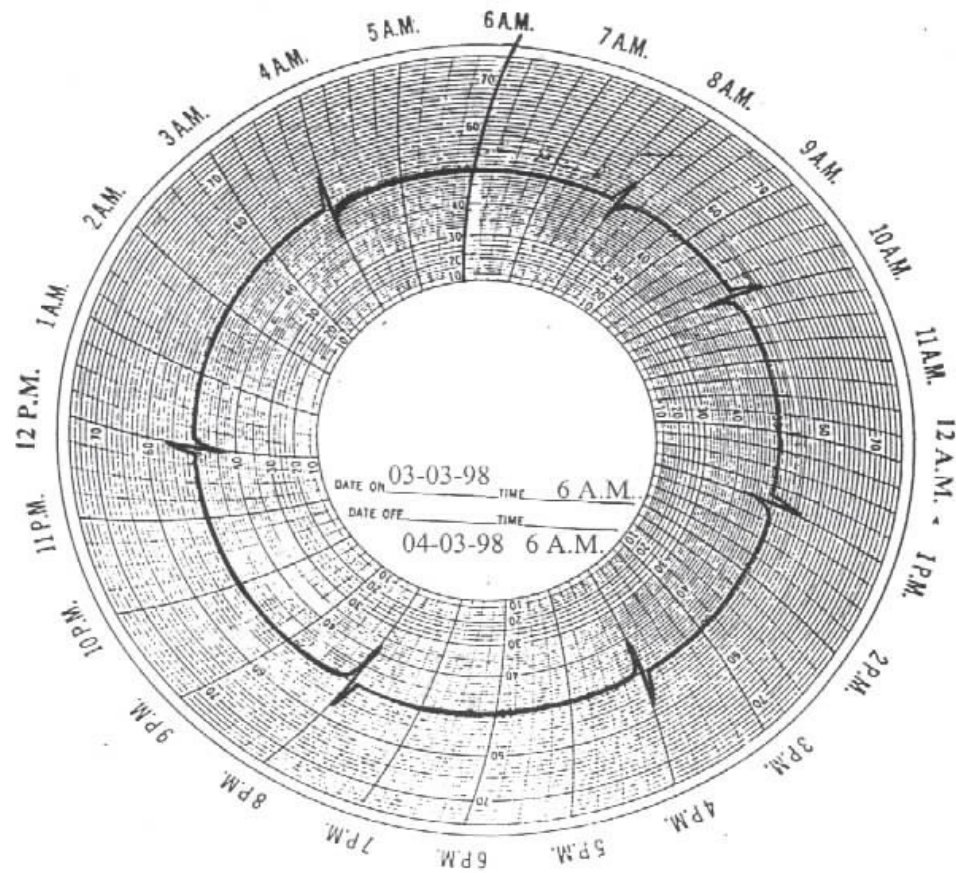


Fig. 2

Pump is operating as normal. However, spikes are noticed, which are presumably due to voltage fluctuation. As a result current fluctuates as spikes, where range of spikes is limited within the overload & underload settings.

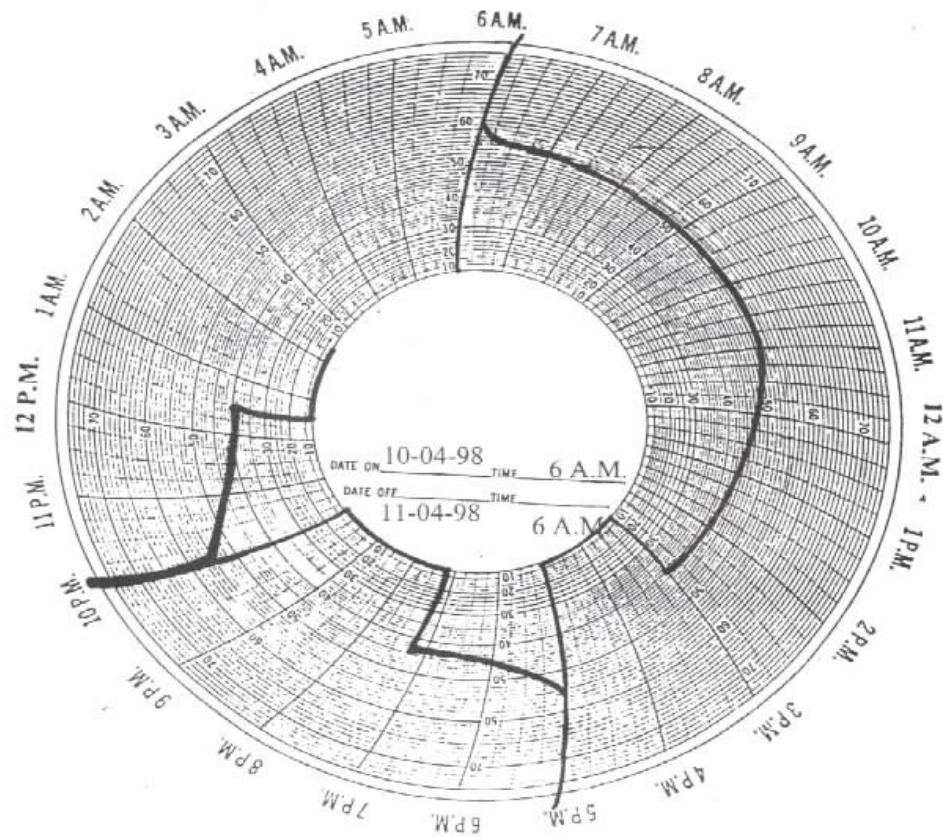


Fig. 3

This is pump off condition of the pump. The discharge of the pump > fluid intake from the wellbore. As A result continuous drop in amperage is observed and as the amperage drops below the underload setting, switchboard automatically switches off the power. Immediately then re-start timer starts working & after a pre-fixed pause time the switchboard automatically switches on the power to ESP.

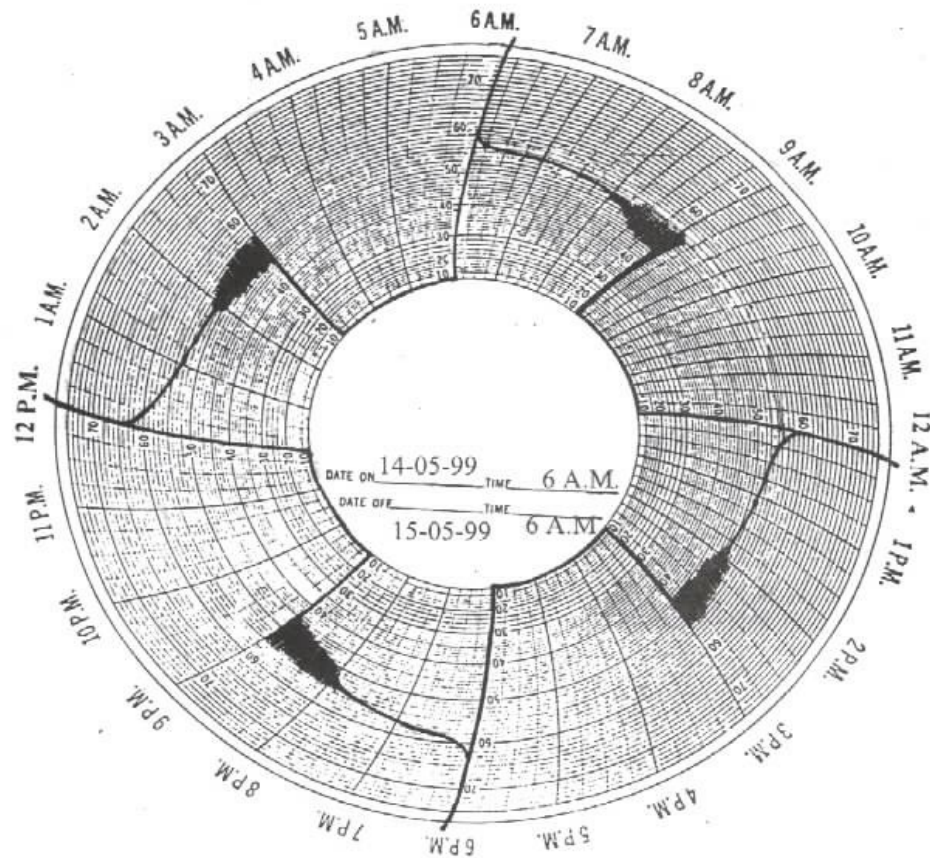


Fig. 4

This is a case of pump off / gas lock. Due to more capacity of pump discharge than the inflow capacity of well, fluid level in the annulus goes down continuously. With generation of more and more free gas in the pump. Finally, the current goes below the underload setting and pump stops. Thereafter the auto restarting after A pre-determined pause. Very frequent current fluctuations prior to shut-down indicates gas locking phenomena.

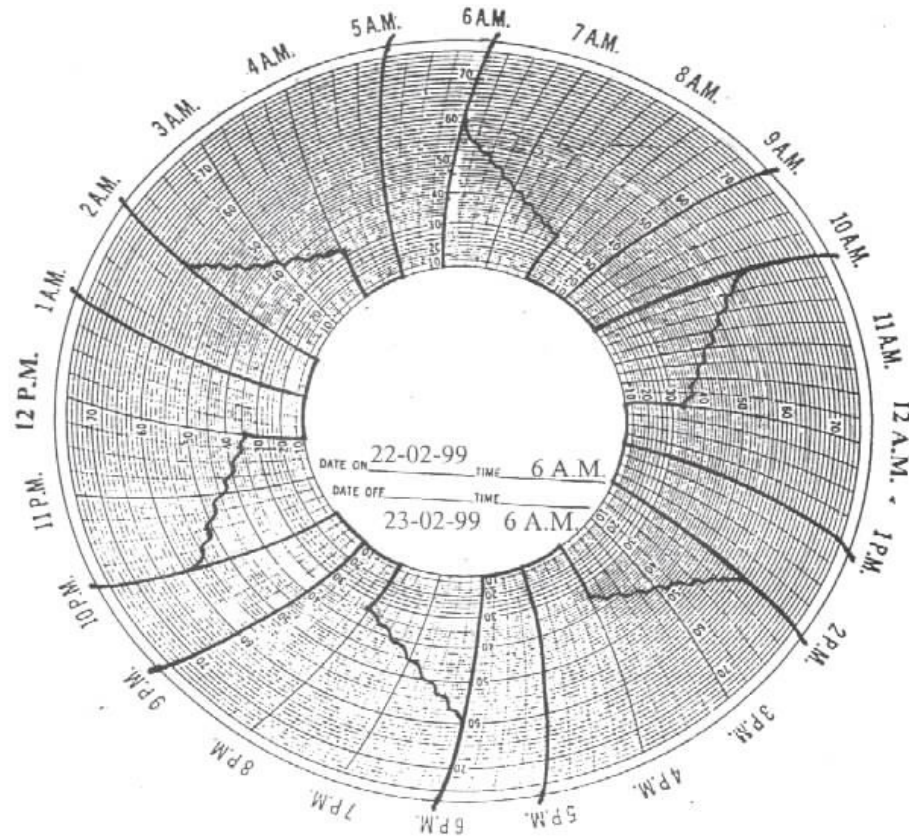


Fig. 5

This indicates insufficient pause time before the pump is restarted. This is possibility due to very low influx into the wellbore & as such length of the pause time should be more. This is also a case of pump-off phenomena.

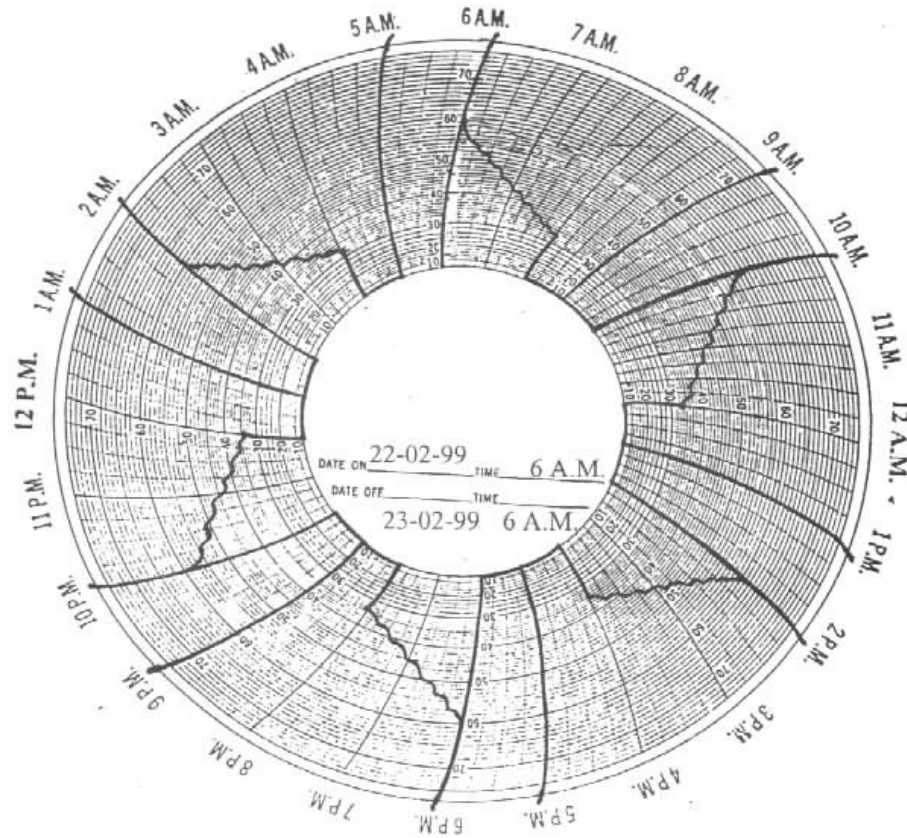


Fig. 6

This indicates insufficient pause time before the pump is restarted. This is possibly due to very low influx into the wellbore & as such length of the pause time should be more. This is also A case of pump off phenomena.

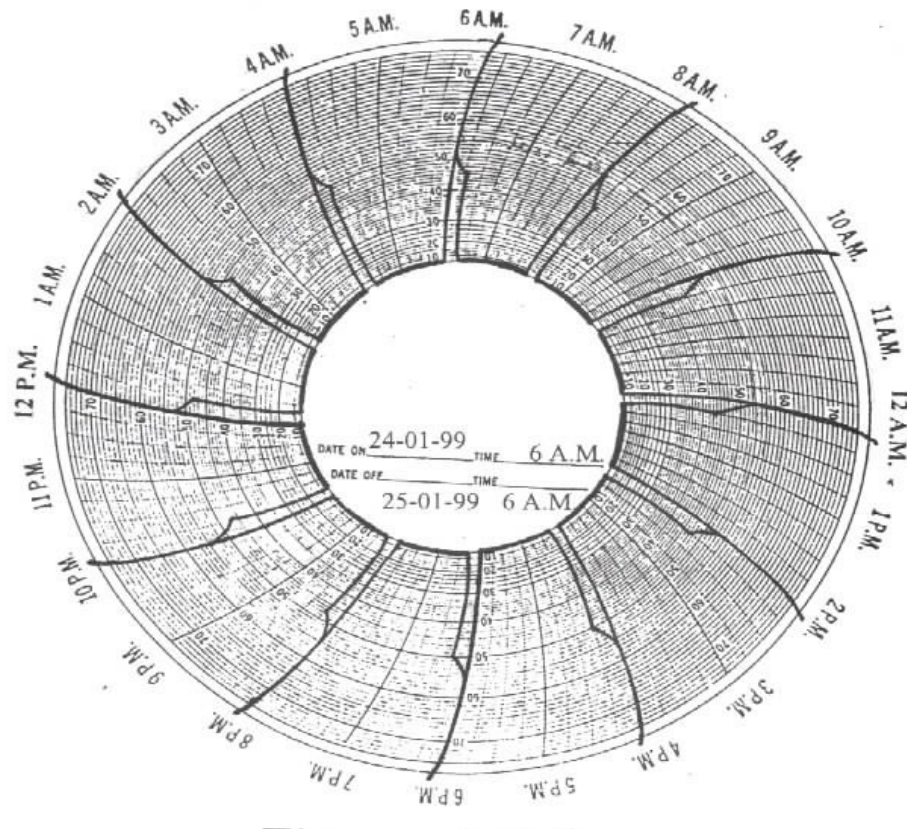


Fig. 7

This indicates A large number times of operation and shutdown of pump. The possible reasons and shutdown of pump. The possible reasons may be :

- = If pump is of very high capacity and inflow into the wellbore is very less. Then this phenomena can occur. However, in practice, except at the initial stages of pump commissioning, this is remote.**
- = If flowline pressure is excessively high it will lead to early shutdown of pump due to underload phenomena.**

After a pre-fixed pause, the automatic restarter starts.

This condition shortens the operating life of pump and hence should be avoided .

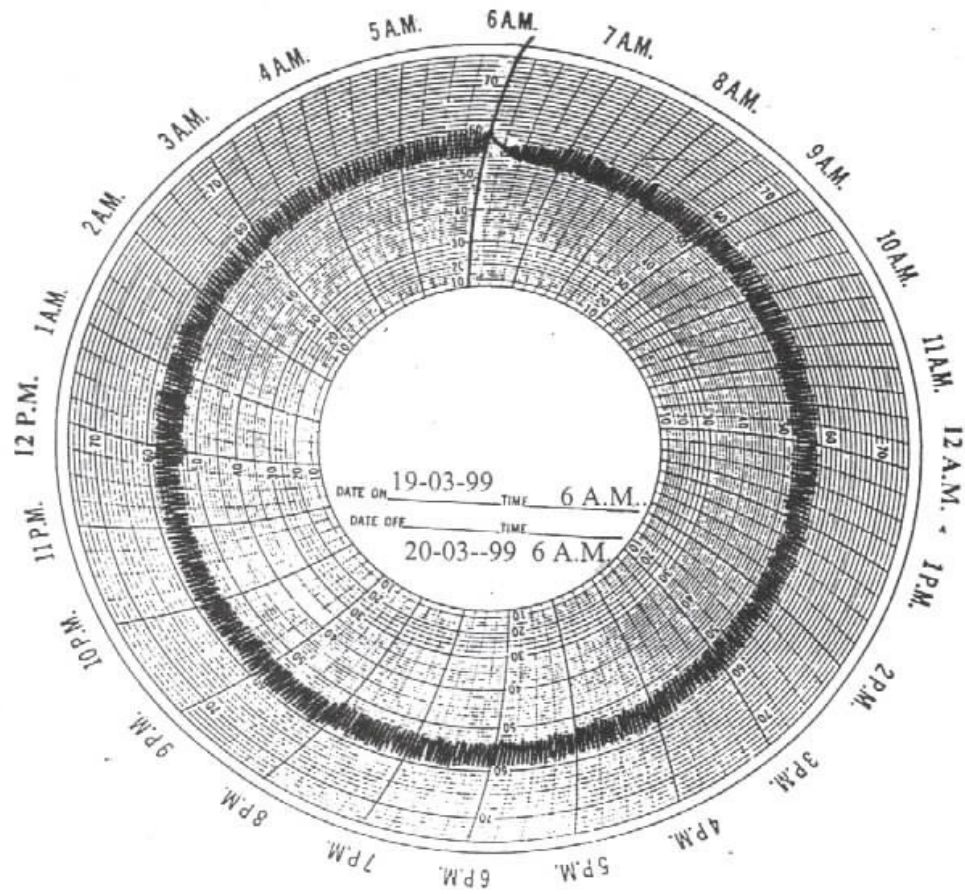


Fig.8

It is a normal / high GOR pump operation, provided there is no voltage fluctuation to that extent. The fluctuation indicates the loading and off-loading of pump very frequently due to interference of gas .

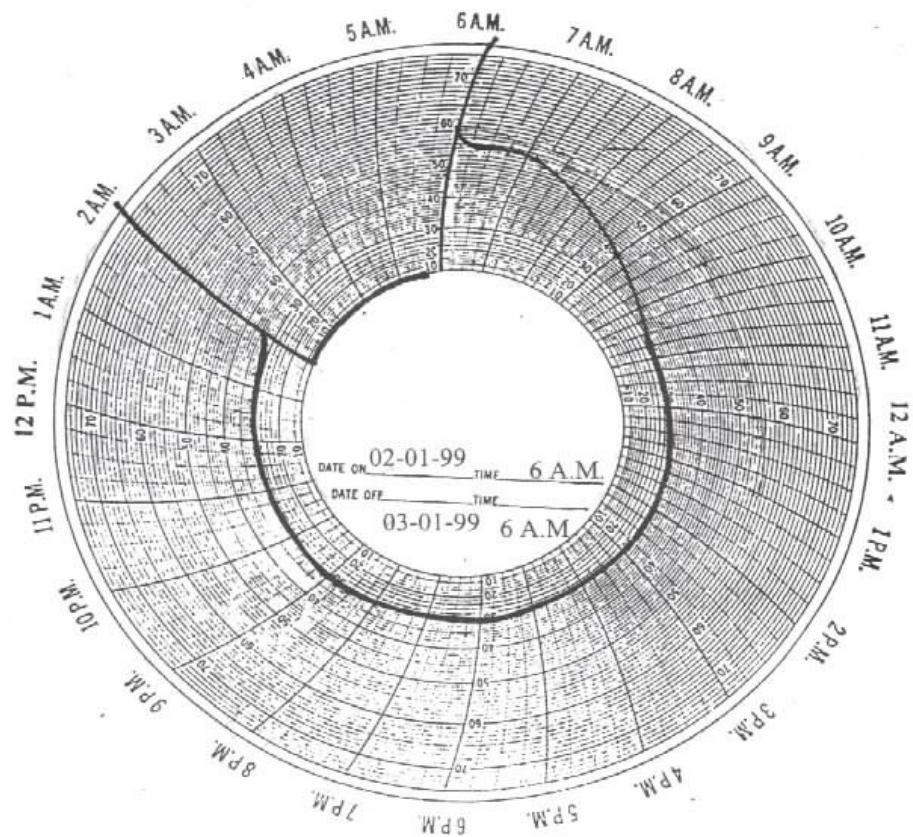


Fig. 9

This is case of pump off / dry run of pump this case arises when underload setting is not done. Due to relatively low intake of formation fluid into the well-bore. Amperage gradually drops & then maintains A very low value all through the pump operation. There is no outflow of fluid at the surface.

This resulted in abnormal heating of the motor / cable and thereby damage the insulation. *This condition must be avoided.*

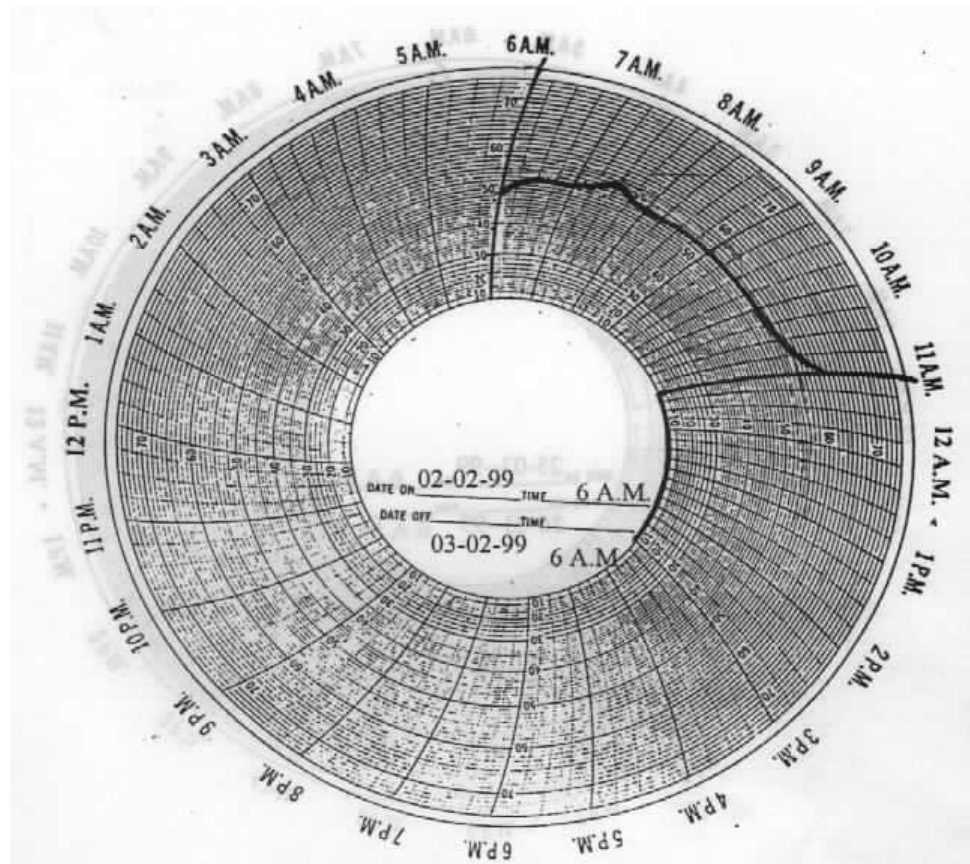


Fig. 10

This is a case of overload. The current goes up gradually & finally when it crosses the overload setting mark. The pump stops. Overloading phenomena may arise due to power fluctuation or due to mechanical / fluid problem. As per convention of the manufacturer, after overload shut-down, pump will not get automatically re-started. This needs to be checked by electrical & production engineers for locating the actual fault. Once the fault is located & rectified, pump is then run manually.

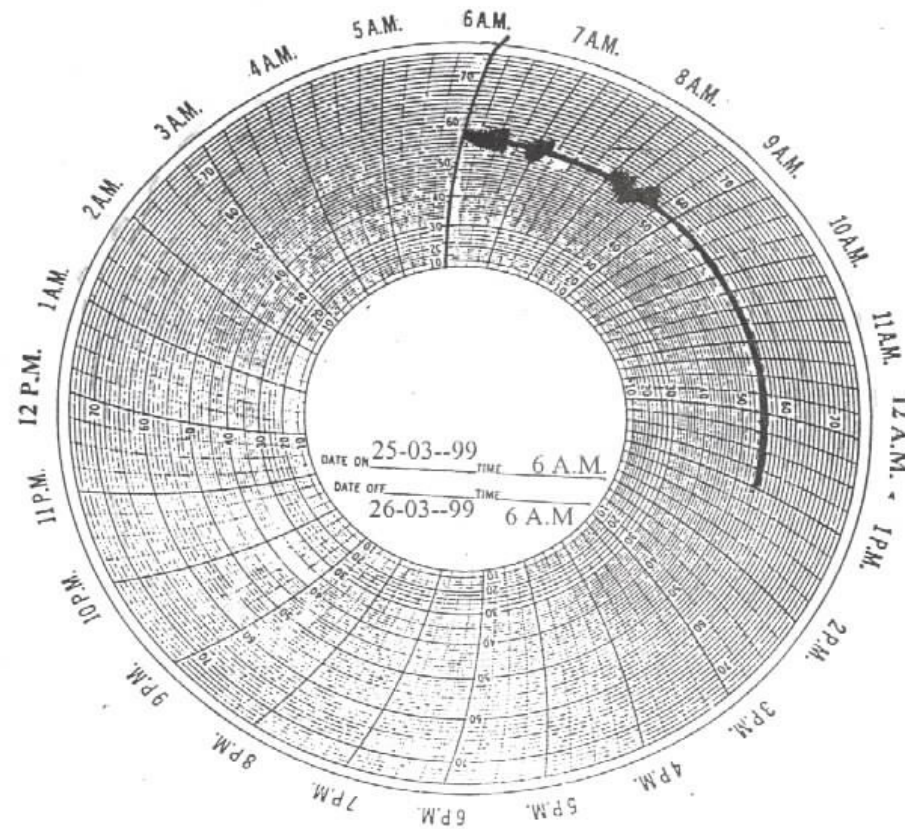


Fig. 11

This is case of viscous / emulsified fluid entry in the pump. Once viscous & emulsified fluid are pumped out, pump is in normal operation.

ESP DESIGN

Given Data

Well depth	: 2500 Mts.. (8220 ft)
Tubing	: 2 7/8 inch
Casing size	: 5 1/2 inch (17 - 20 ppf)
Reservoir pressure	: 205 kg/cm ² (2915 psi)
Flowing Bottom hole pressure	: 180 kg/cm ² (2560 psi)
Wellhead pressure	: 7 kg/cm ² (100 psi)
Water cut	: 60%
GOR	: 45 m ³ /m ³ (252 SCF/b)
GLR	: 17 m ³ /m ³ (96 SCF/b)
Design Liquid rate (at stock tank)	: 35 M ³ /D (220 b/d).

ESP DESIGN

Degree API of oil	: 35° API (Sp.Gr. = 0.8489)
Specific gravity of gas (Air = 1)	: 0.65
Specific gravity of water	: 1.05
Bottom hole Temperature	: 70°C (158°F)
Wellhead temperature	: 30°C (86°F)
Bubble point pressure	: 80 kg/cm ² (1137 psi)
Electric supply system	: 400 / 440 V : 50 Hz
Well profile	: S - shaped (Build-up and
Build-down	Profile) and from 1000 Mts. depth from surface it

is vertical.

Formation volume factor (B_o)	: 1.15
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The following are the step-wise calculations :-

STEP - 1 : SIZE OF PUMP

From the catalogue of ESP manufacturer the best suited pump primarily with respect to its OD and capacity is to be selected. Let the available ESP is of REDA make.

Since casing size is 5 1/2", at the first instance, 400/450 series REDA pump/protector as applicable in 5 1/2", is considered (Reference : REDA catalogue). Now, maximum OD of Reda pump set with cable, cable guard and cable clamp in position is required to be checked with 5 1/2" ; 20 ppf casing (that is minimum ID of casing).

STEP - 1 : SIZE OF PUMP

- i) OD of 450 series protector = 114.3 mm
 - ii) Thickness of Armoured cable of 6 AWG of parallel shape = 12.3 mm
 - iii) Thickness of cable guard and cable clamp = 2.0 mm (approx..)
-

Total = Max. OD of REDA Pump (400 / 450 series) = 128.6 mm

I.D of 5 1/2" , 20 ppf casing = 121.4 mm

Drift diameter of 5 1/2", 20 ppf casing = 118.2 mm

Since drift diameter of 5 1/2", 20 ppf casing is less than Max. O.D of Reda pump, it is required to find out pump of one size lower.

STEP - 1 : SIZE OF PUMP

the next lower size is of 338/325 series pump / protector as applicable in 4 1/2" casing.

i) OD 338/325 series pump / protector = 85.85 mm

ii) Thickness off Armoured cable = 12.30 mm
(6 AWG, parallel)

iii) Thickness of cable guard and cable clamp = 2.00 mm

Total = Max. OD of 338/325 series Pump / protector = 100.15 mm

Therefore, clearance between minimum casing I.D. and max. pump O.D.

$$\begin{aligned} &= (118.20 - 100.15) \times 1/2 \\ &= 18.05 \times 1/2 = 9.0 \text{ mm} \end{aligned}$$

From Reda catalogue, the compatible pump / protector set of 338/325 series is selected which is to be coupled with 375

STEP - 2 : STATIC AND DYNAMIC LEVEL

Considering the datum level at 2500 Mts., and with specific gravity of water as 1.05, the fluid level at static condition = $2050 \times 1/1.05 = 1952$ Mts.. So, static fluid level from the surface = $2500 - 1952 = 548$ Mts..

The fluid level at flowing condition (that is, dynamic condition) = $1800 \times 1/1.05 = 1715$ Mts..

Therefore, dynamic level from the surface

$$= 2500 - 1715 = 785 \text{ Mts..}$$

STEP - 3 : LOCATION OF PUMP DEPTH

The pump has to be located below the dynamic. Also, to minimise the interference of free gas, the pump, if possible, can be located in deeper depth.

i) Dynamic level from surface = 785 Mts..

ii) Bubble point pressure of 80 kg/cm²,
Which is equivalent to = 762 Mts..

Total = 1547 Mts..

∴ Location of Pump = 1600 Mts.. from surface

STEP - 4 : FLUID VOLUME IN THE PUMP (Q)

$$\begin{aligned} Q &= 35 \text{ m}^3/\text{d} \times B_o = 35 \times 1.15 \\ &= \boxed{40 \text{ M}^3/\text{d}} \end{aligned}$$

STEP - 5 : PUMP SELECTION

A 400 pump is selected from performance curve as supplied by the manufacturer for 50 Hz supply and 338 series pump, here the desired fluid production rate of 40 m³/d lies in the recommended range for operating the pump on the accepted efficiency level.

STEP - 6 : PUMP STAGES CALCULATION

From the performance curve, 100 stage develop 400 Mts. of head.

∴ 1 stage develop $400/100 = 4$ Mts. of head.

Now, TOTAL Head required, that is, total dynamic head (TDH) will be,

TDH = Dynamic level from surface + fluid friction in the tubing + Tubing head Pressure.

= 785 Mts. + negligible + 70 Mts..

= 855 Mts. 855 Mts.

∴ Total stages of pump required = $\frac{855 \text{ Mts.}}{4 \text{ Mts. / stage}} = 214 \text{ stages.}$

From the catalogue of the manufacturer, the number of stages and housings have been selected, so that total stags of pump is slightly more or equal to 214 stags.

2 numbers of housings each having 81 stags and 1 number having 60 stages have been selected.

So, total stags = $(2 \times 81) + 60 = 162 + 60 =$ 222 stages

STEP - 7 : MOTOR HORSEPOWER REQUIREMENT

From performance curve, max. H.P. = 6.0 H.P. / 100 stages

The nearest whole number = 6.0 H.P./100 stages

= 0.06 H.P. / stages

So, Total H.P. requirement = H.P. /stages x Number
of Stage x specific
gravity of water

= 0.06 x 222 x 1.05

= 13.98 H.P.

ESP DESIGN

STEP - 7 : MOTOR HORSEPOWER REQUIREMENT

From catalogue, 375 series motor has to be selected, which has H.P. either equal to this valve or next higher valve.

H.P. Motor selected = 16.3 H.P.

It is always advisable to choose a motor with low amperage rating, provided its voltage rating is not very excessive. So, from two categories of 16.3 H.P., 50 Hertz motors,

That is from, 16.3 HP; 238 V ; 38 A; 50 Hertz

and 16.3 HP; 323 V; 25 A ; 50 Hertz

the motor of 16.3 H.P.; 323 V; 25 A; 50 Hertz is selected.

ESP DESIGN

STEP - 8 : MAIN CABLE SELECTION

From manufacturers catalogue "Redelene" type (can work up to 205° F, where B.H.T. is 158°F) flat cable and 4 AWG (considering cost and voltage drop factor) has been considered.

ESP DESIGN

STEP - 9 : SURFACE VOLTAGE CALCULATION

From cable voltage chart, supplied by the manufacturer,

Voltage drop = 11 volt / 1000 ft.

Total cable length = Subsurface cable length +
Surface cable length
= 1600 Mts.. + 100 Mts.. (say)

Total cable length = 1700 Mts.. = 5576 ft
= 5600 ft

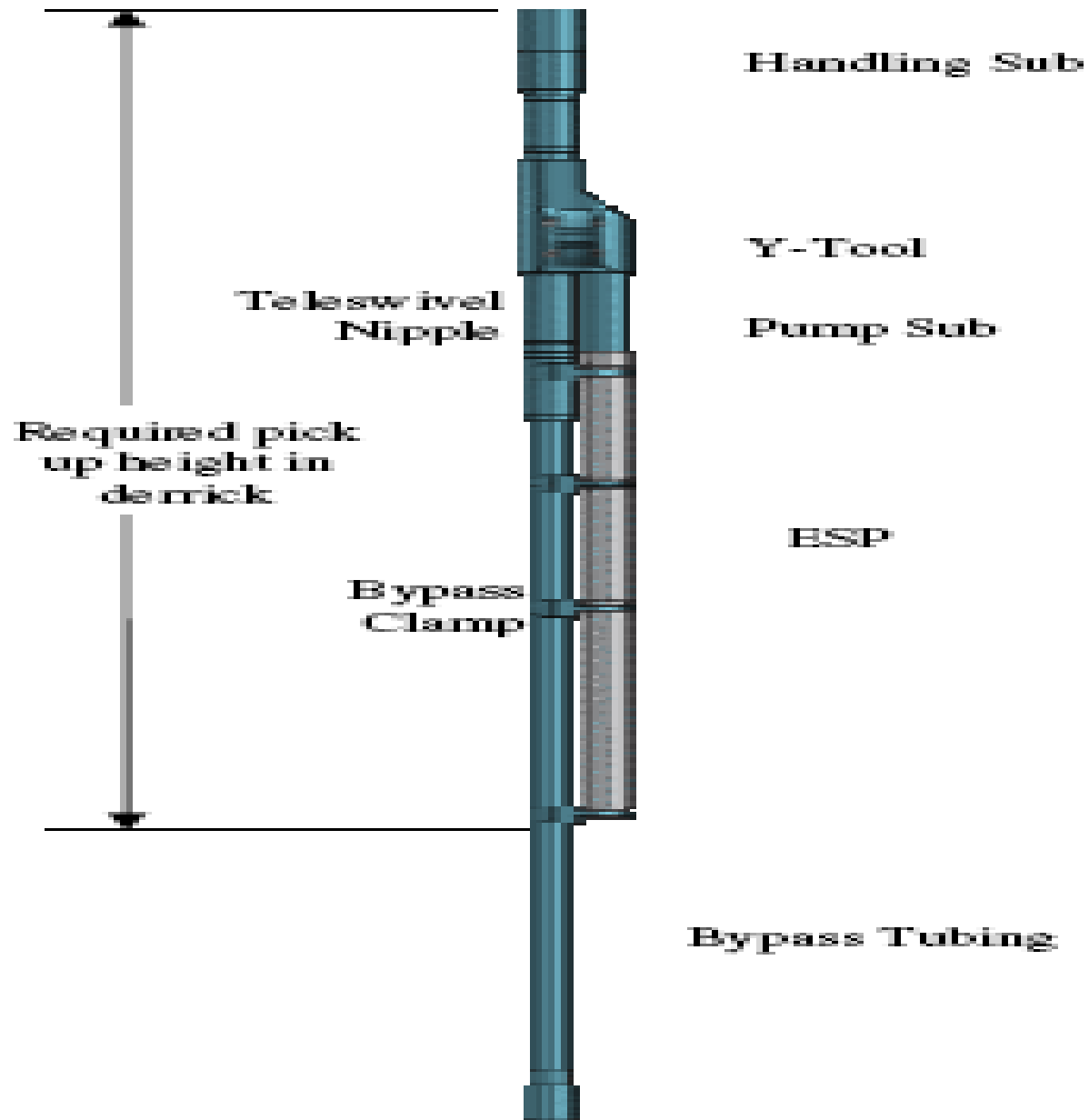
So, the total voltage drop = $11\text{V}/1000\text{ft} \times 5600 \text{ ft.}$
= 61.6 v

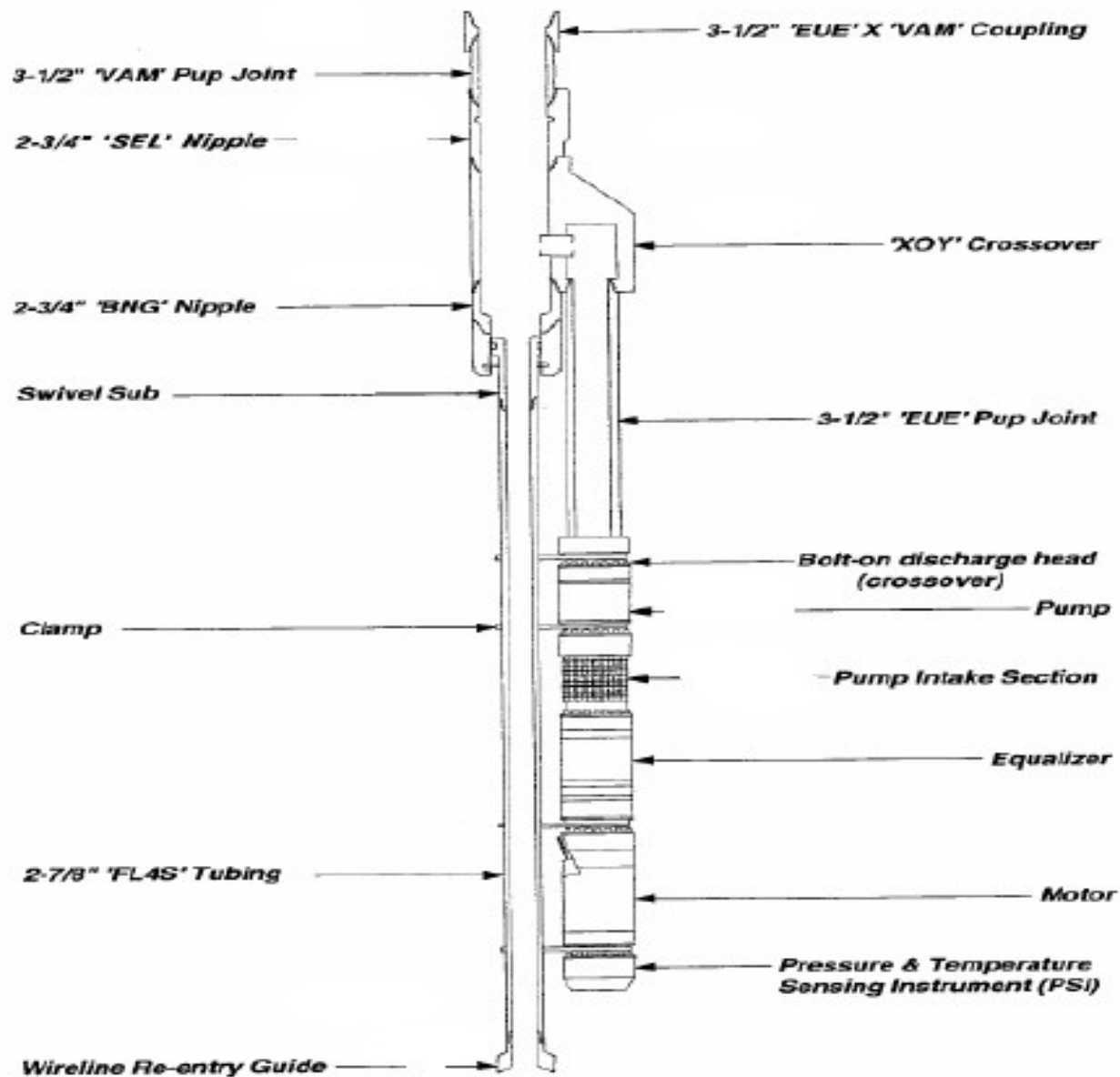
∴ Voltage required at the surface = name plate voltage +

Total Voltage drop = 323 + 61.6

= 384.6 V

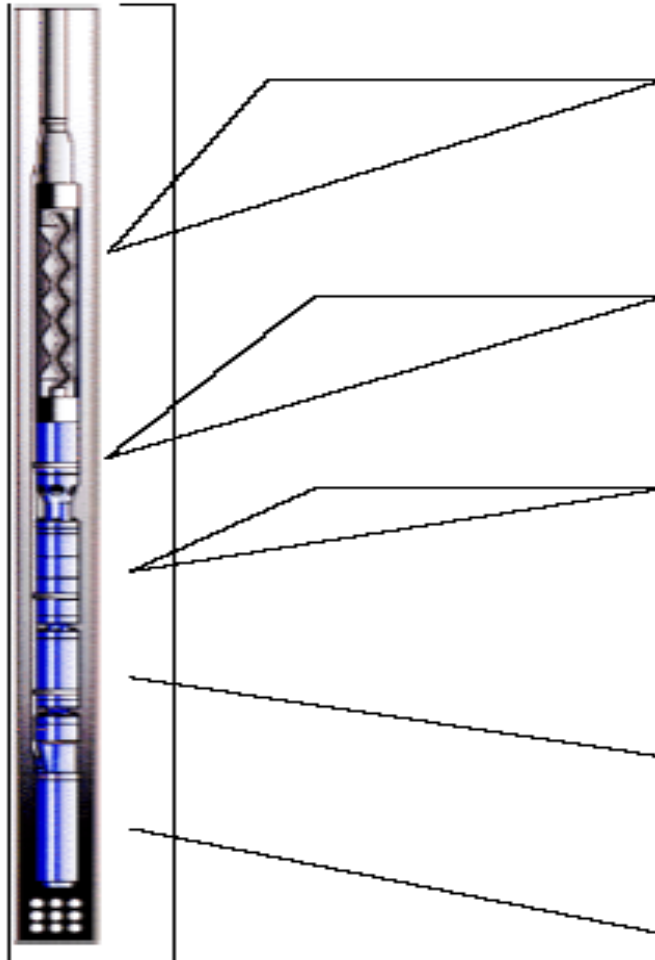
= 385 V





Typical Field ESP installation with Y-tool bypass to allow access to the reservoir below the pump.

ESPCP



Progressing Cavity Pump

High efficiency pump for handling high viscosity fluids and sand.

Flexshaft & Pump Intake

These components convert the eccentric motion of the PCP pump to concentric motion necessary for the seal gear reducer & motor.

Seal Section

The seal section isolates the motor & gear reducer oil from the well fluid. It also absorbs the thrust load generated by the pump as well as allowing for the expansion & contraction of the system oil due to heat.

Gear Reducer

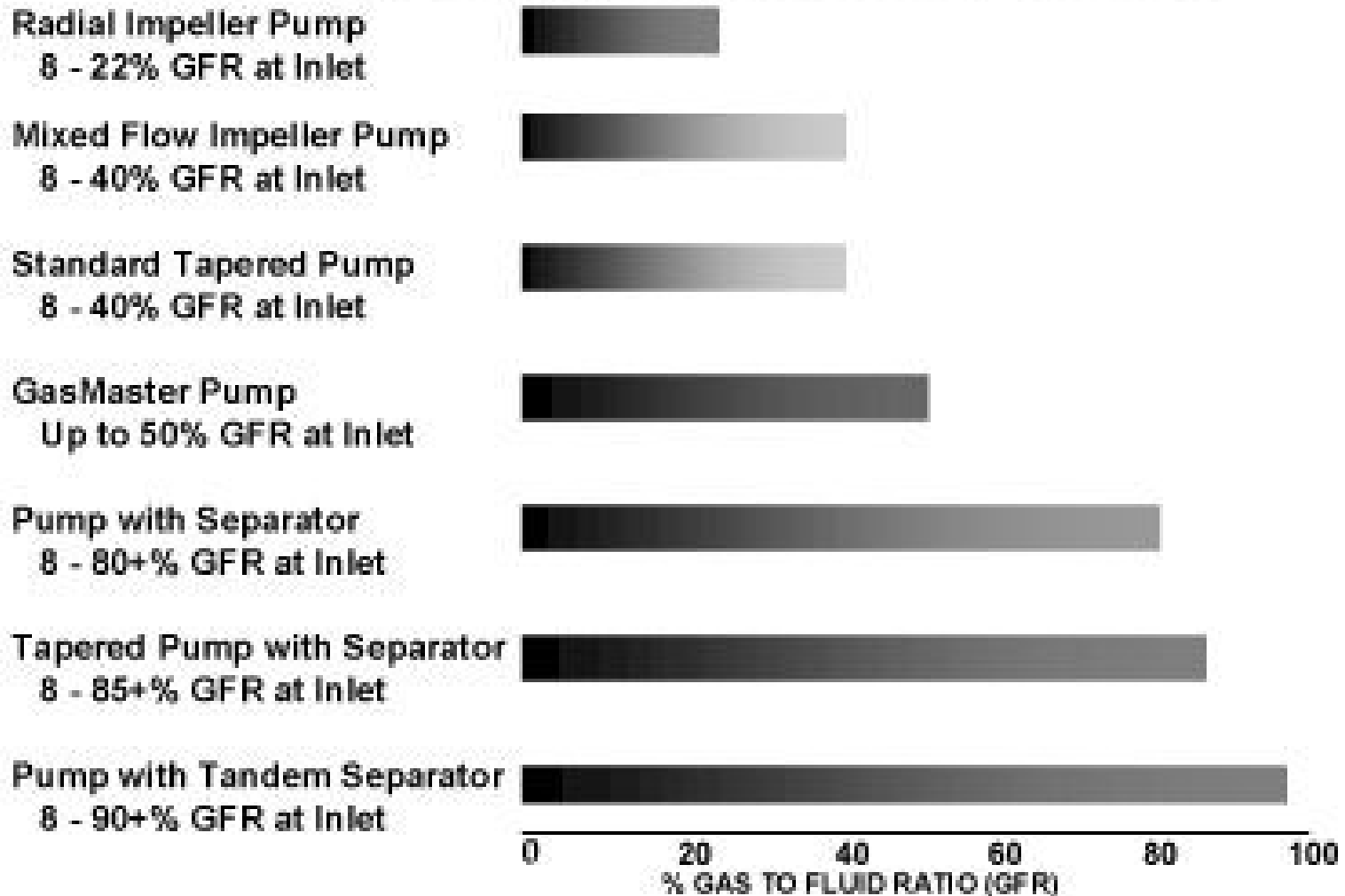
The gear reducer reduces the motor speed (3500 RPM @ 60 Hz) to speeds acceptable to the PC Pump.

Electric Motor

This utilizes a standard electric submersible motor as used on ESP systems.

ESP in high GOR wells

ESP CAPABILITIES IN GASSY CRUDE



Advantages.

- ➡ Reduction of power requirements
- ➡ High volumetric efficiency
- ➡ Handling of solids and abrasive particles
- ➡ Handling viscous fluids
- ➡ Does not create emulsions
- ➡ Reduces “Gas Locking” problems

Advantage

- Additionally, it provides the following improvements that are unique to the ESPCP system:
 - ➡ Elimination of rod and tubing problems
 - ➡ Eliminates stuffing box leakage
 - ➡ High torque capacity
 - ➡ Reverse flow capability at low speeds to flush sand
 - ➡ Reduction of Bottom Hole viscosity due to exchange of heat from the downhole motor to the fluid

Installation procedures

- The pumping rig should be centred all over the wells as closely as possible. A guide wheel, cable sheave should be secured safely to the rig mast no higher than 30 to 45 ft above the wellhead. The guide wheel should be at least 54 in. in diameter .
- The cable reel or spooling truck should be positioned about 100 ft from the wellhead in direct line of sight of the rig operator

- Make certain that the proper scale ammeter chart paper is on the recorder, that the pen is operating properly and that the setting for the day and time are correct.
- Electrical checks phase-to ground and phase to –phase should be made prior to start-up and readings recorded. Phase to phase readings must be balanced

- With all checks completed, start the equipment .
- Immediately after start-up, check the line current with a “clamp-on” ammeter and record. Using this information, calibrate the recording ammeter

- If actual “pump up” time is exceeding calculated “pump up “ time, it should be assumed that the pump is in reverse rotation and appropriate action taken.
- Liquid level in the annulus need to be checked constantly during initial running of pump so as to establish the stable working fluid level .

- The pumping rig should be centred all over the wells as closely as possible. A guide wheel, cable sheave should be secured safely to the rig mast no higher than 30 to 45 ft above the wellhead. The guide wheel should be at least 54 in. in diameter

Thank you

